Section 1

Networking 2 - Final Project: Infrastructure Design & Simulation

Course: Networking 2
Professor: Sir James Dean Valeza San Miguel
Project Title: 7-Eleven Network Infrastructure Design
Deadline: May 20,2026(Wednesday)

Section 1: Project Overview

Chosen Organization:

7-Eleven Philippines – A leading convenience store chain operating 24/7, providing retail goods, fresh food, beverages, and essential services to customers.

Project Scope:

This project involves designing, documenting, and simulating a complete network infrastructure for 8 branches of 7-Eleven Philippines in Valenzuela City using Cisco Packet Tracer. The solution will implement VLSM-based IP addressing with a minimum of 20 subnets, VLAN segmentation with Router-on-a-Stick for inter-VLAN routing, centralized DHCP with relay, DNS, FTP, HTTP, and VoIP services, dynamic routing OSPFwith default routing for internet connectivity, Link Aggregation (LAG) for uplink redundancy, and hardened device configurations—all supported by comprehensive documentation including floor plans, rack diagrams, cable labeling schemes, a Bill of Materials with PHP pricing, and OSI Model mapping to meet all course requirements.

Branch Locations:

1. **Branch (Gen T.)#4473** – Gen. T. de Leon, Valenzuela, 1442 Metro Manila
2. **Branch(Ugong 1) #6039** – Gen. T. de Leon, Valenzuela, Metro Manila
3. **Branch (Lorex Ave)** – Lorex Avenue, Valenzuela, Metro Manila
4. **Branch (Parada)#95** – Capt. Cruz, Valenzuela, 1443 Metro Manila
5. **Branch (Dionisio Alley)** – 49 Dionisio Alley, Valenzuela, Metro Manila
6. **Branch (ARCA Plaza)** – Unit 7 & 8, ARCA Plaza Commercial Building, 73 Maysan Rd, Maysan, Valenzuela, 1440 Metro Manila
7. **Branch (C.J. Santos)** – C.J. Santos Cor. Maysan Rd, Valenzuela, 1440 Metro Manila
8. **Branch (12th Road)** – 12 Road 1, Valenzuela, Metro Manila

Organization Size:

- **Total Branches:** 8 locations
- **Store Size:** Approximately 2,400–3,200 square feet per branch
- **Staff per Branch:** 4–8 employees (mix of cashiers, store crew, and branch supervisor)
- **Total Network Endpoints:** ~150+ devices across all branches (POS terminals, CCTV cameras, PCs, IP phones, servers, and network equipment)

Functional Areas (per branch):

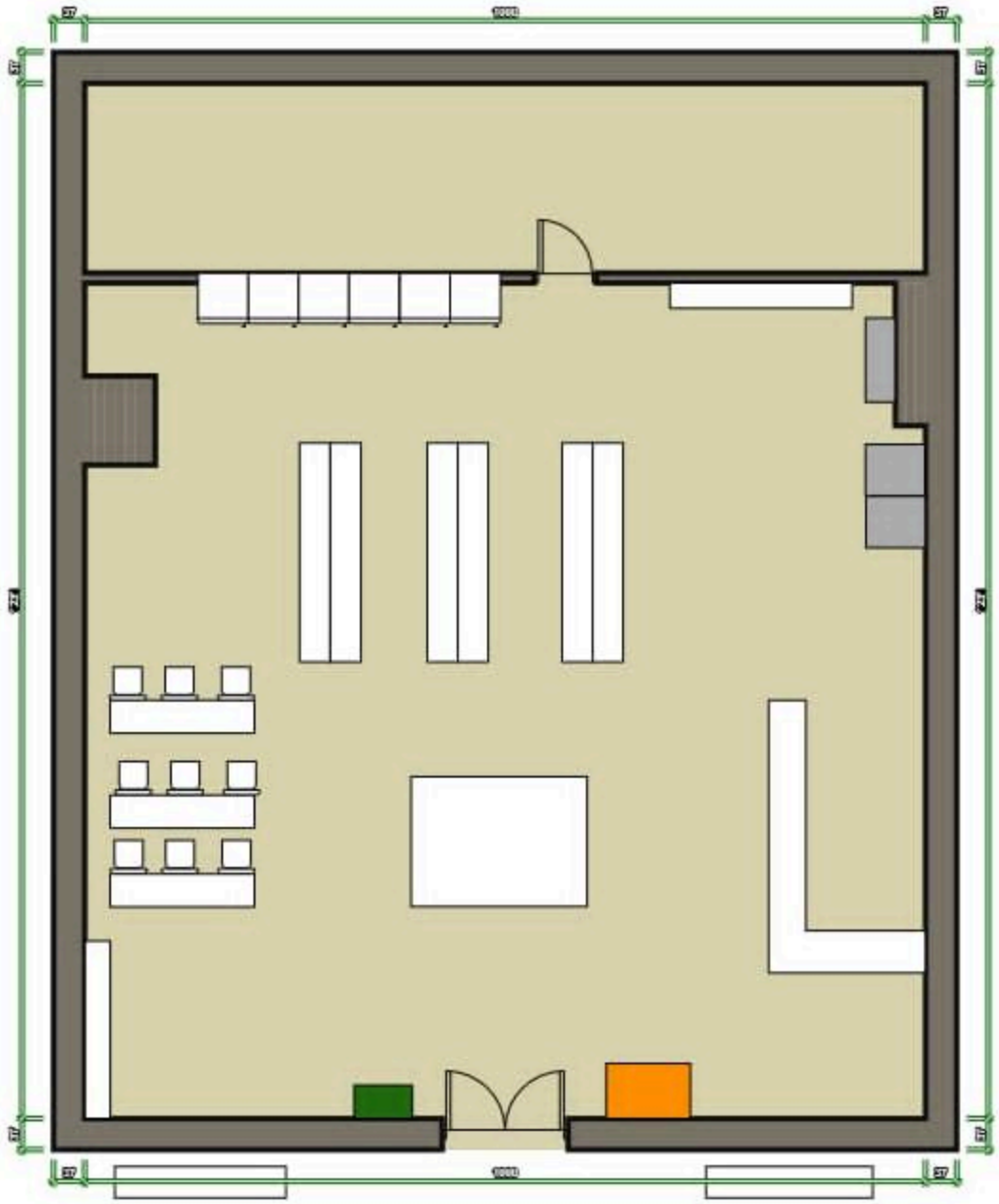
- **Sales Floor** – Customer shopping area with product shelves and refrigerated displays
- **Checkout Counter** – Point-of-Sale (POS) terminals and cashier stations
- **Food Service Area** – Hot food preparation and Slurpee station
- **Customer Dining Area** – Seating area for in-store consumption
- **Back Office** – Administrative workstations, inventory management, and staff area
- **Storage/Warehouse** – Inventory and supplies storage
- **Server/Network Room (MDF)** – Central network equipment, servers, and telecommunications

Number of Floors:

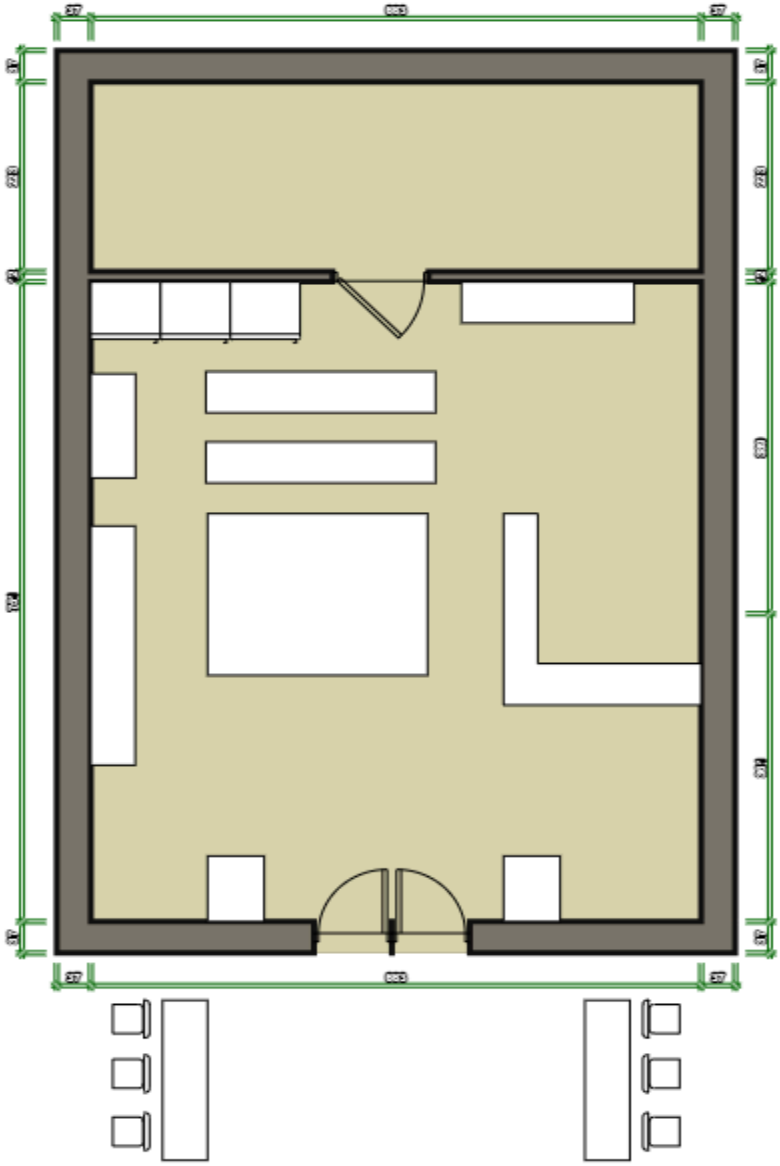
Single-story layout per branch (ground floor only)

Logical Floor Plan:

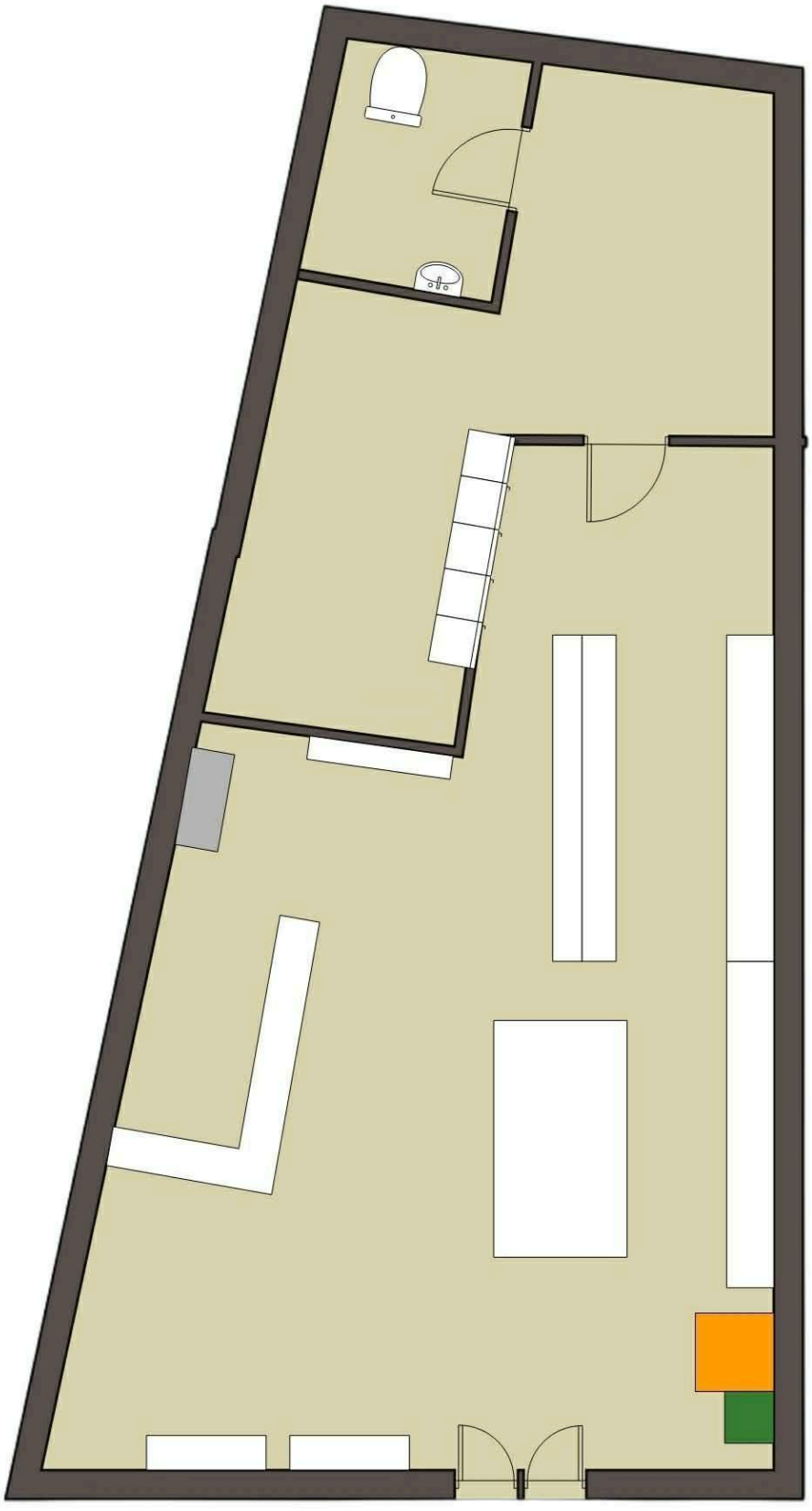
12 Road Branch



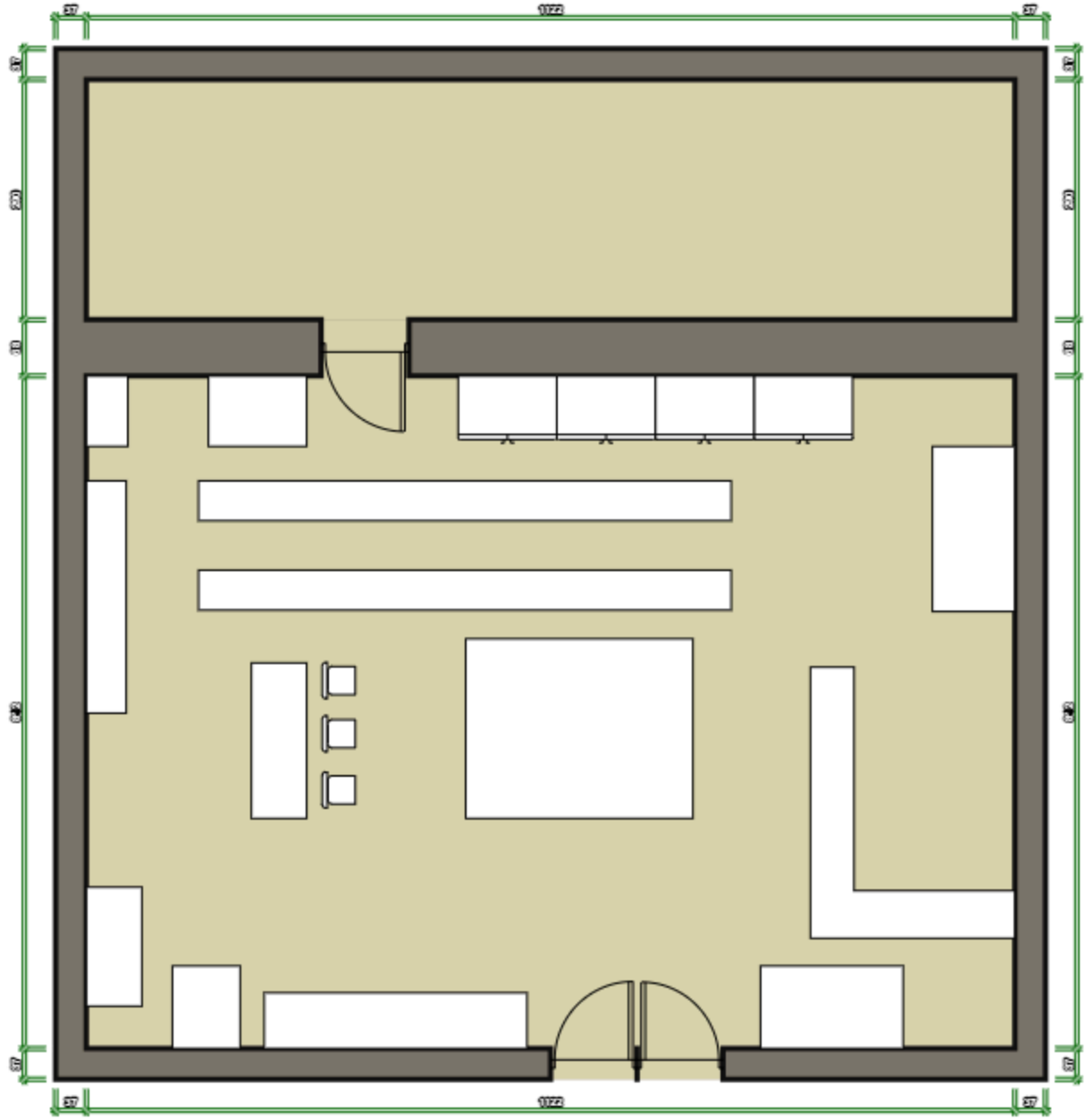
Maysan Branch



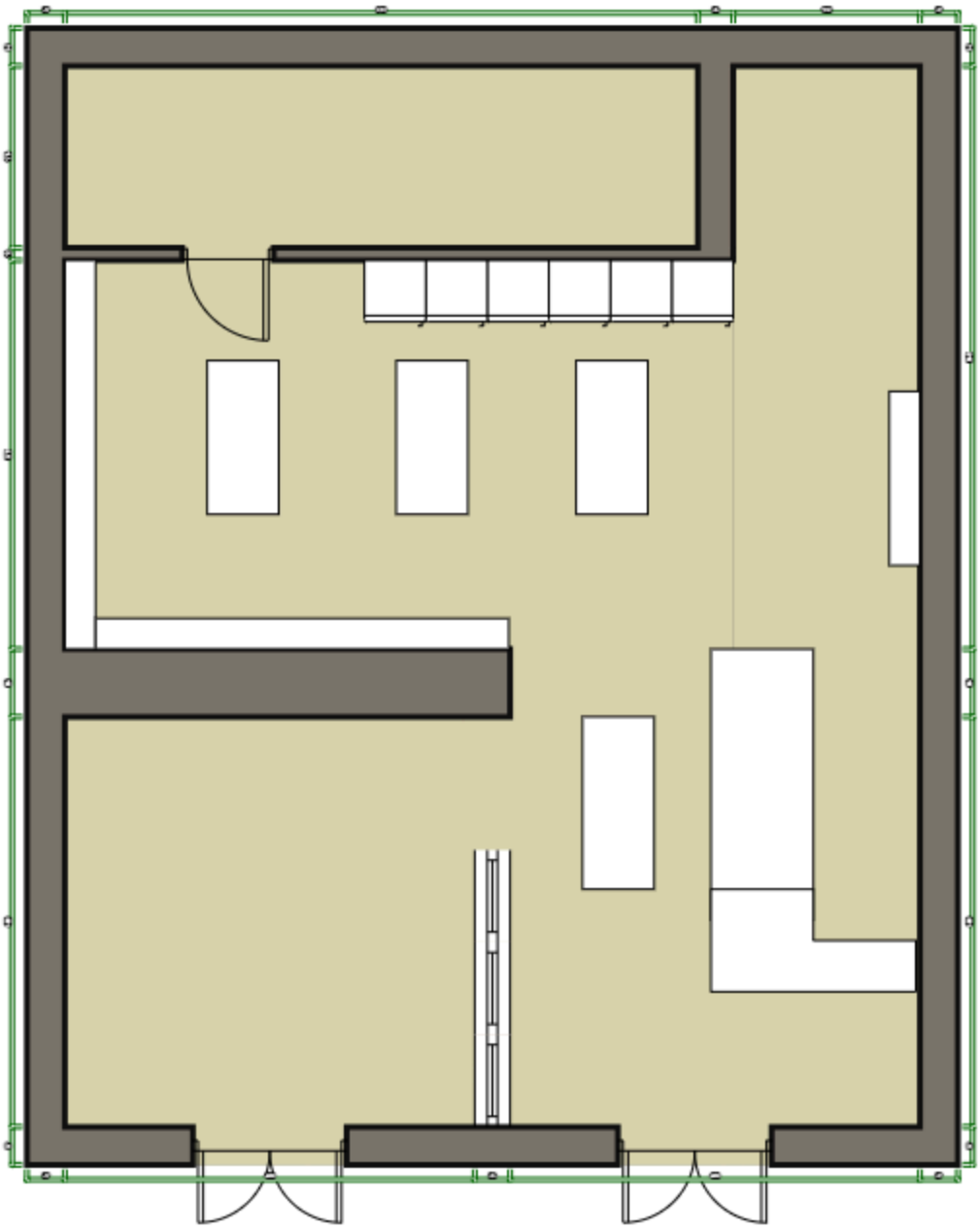
Gen T. Branch



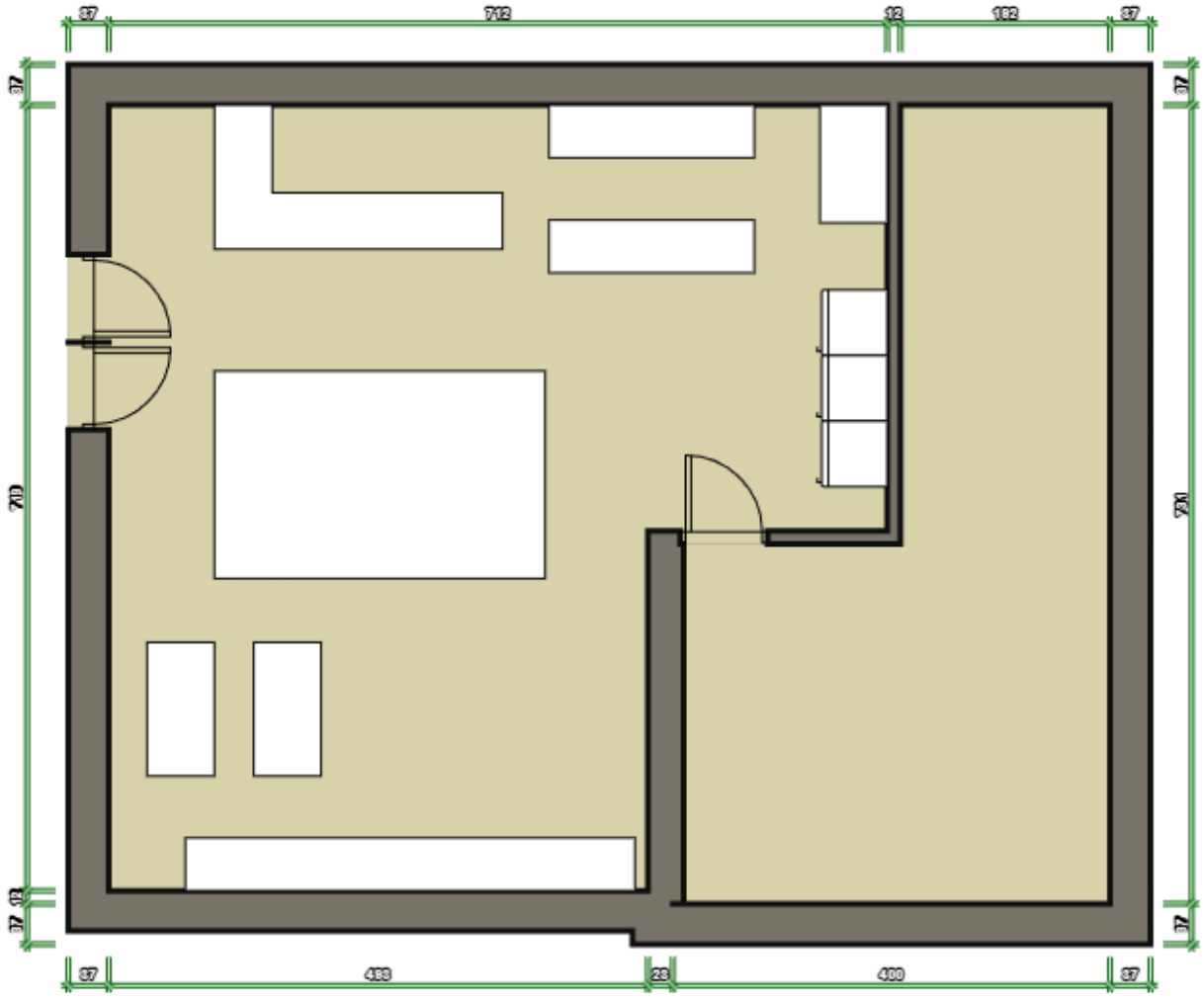
Malinta Branch



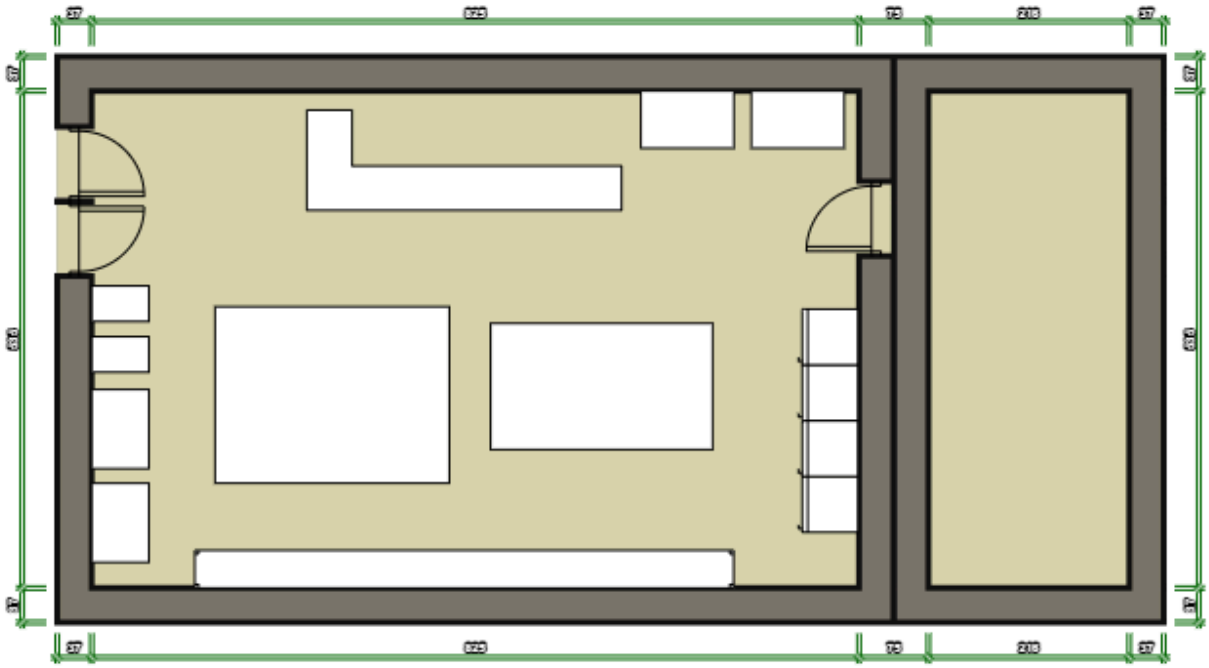
Maysan Branch Arca Plaza



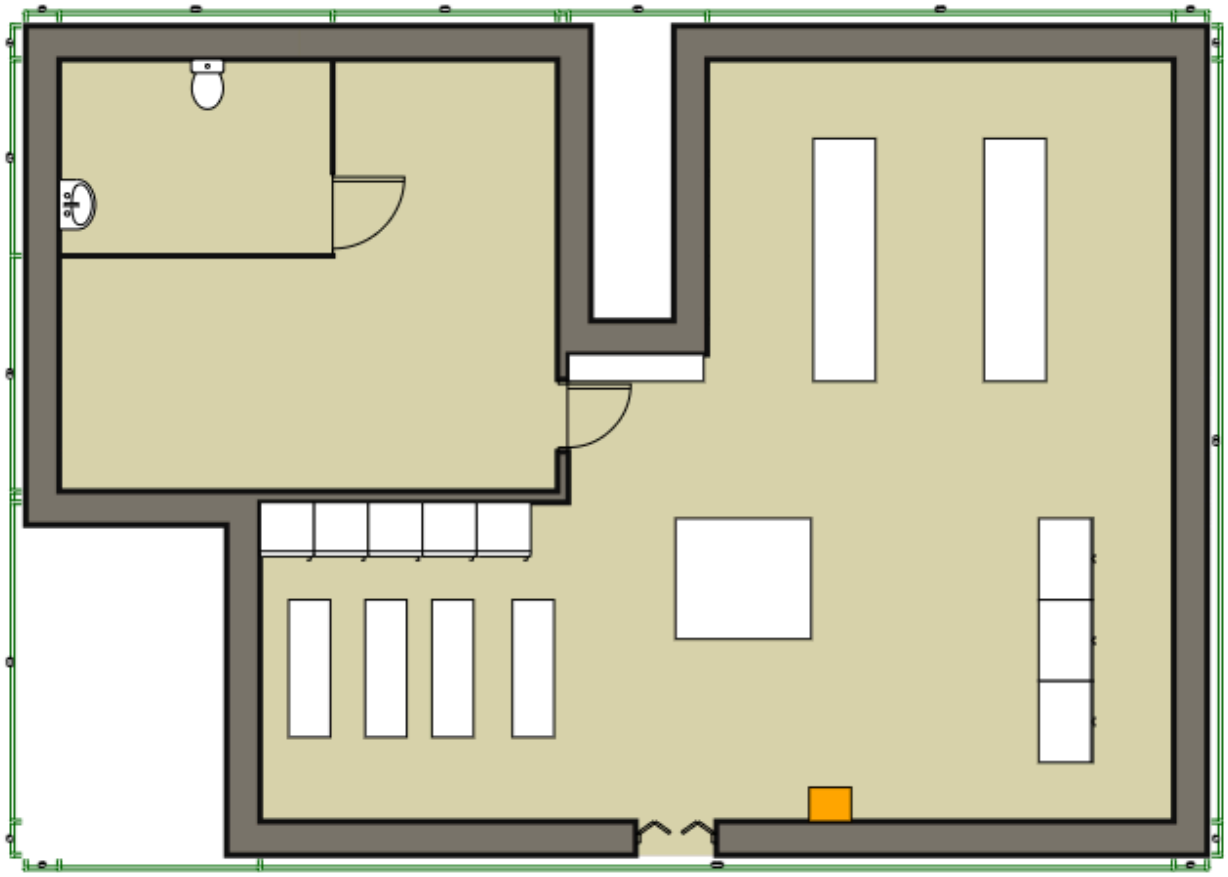
Parada Branch



Lorex Branch



Ugong Branch



Section 2 Bill of Materials

Section 2: Bill of Materials & Device Documentation

Total Estimated Budget: ₱5,977,372

Component Type	Image	Device Model	Unit	Est. Price (PHP)	Purpose
Internet Service Provider					
PC		Dell Vostro 3910 (i3/8GB/256 GB SSD)	8	₱40,00(ea.)	Back-office tasks & admin work
DHCP SERVER		Dell PowerEdge R740/R750	1	₱300,000	Assigns IP addresses & network services
POS		Posiflex XT-3815 G2 (15" Touch)	16	₱45,000 (ea.)	Process Sales & Payments
CCTV Camera		Hikvision DS-2CD214 3G2-IU (4MP)	30	₱5,800 (ea.)	Video Surveillance & Security Monitoring
ATM Terminal		NCR SelfServ 8288 (Bank-Partner)	8	₱350,000 (ea.)	Cash withdrawal & banking services
Kiosk		Posiflex KS-6615 (15.6" Touch)	8	₱95,000 (ea.)	Customer to Self services & buy prepaid load

Router		MikroTik RB4011iGS+ RM	8	₱12,999(ea.)	Internet gateway & Branch Security
Switch		TP-Link TL-SG1428 PE	16	₱14,985(ea.)	Connects & Power Network Devices
Wireless Access Point		Ubiquiti UniFi U6-Pro	8	₱13,250(ea.)	Provides Wireless Internet Access
Telephony Phone		Grandstream GXP1625 (2-line PoE)	8	₱3,800 (ea.)	Voice Calls & Communications
Cabling		CAT6 300m	8	₱2,500 (ea.)	Connects Devices to the network
Patch Panels		AMPCOM Patch Panel, Upgraded Version CAT5E/CaT 6 24 Ports	8	₱3,550(ea.)	Organize Cable Connections
Faceplate		Dual-Gang Wall Faceplate	23	₱30(ea.)	Mounts keystone jack on wall
Keystone Jack		Cat6 UTP Keystone Jack (T568A/B)	46	₱80(ea.)	Terminates cable at wall outlet
Copper Wall Mount		Surface-Mount, Brushed Copper	23	₱150(ea.)	Houses faceplate & jack; protects wall termination

Patch Cables & Accessories		RJ45, Trunking, Labels, Cable Ties	1	₱85,000	Cable termination, organization & protection
UPS		APC UPS 650VA-360W Uninterruptible Power Supply (BVX650I-PH, Easy UPS, 4 Outlet, AVR/Surge)	24	₱3,750 (ea.)	Backup power & surge protection for critical devices
Server Rack		StarTech 42U Server Cabinet (600x1000 mm)	8	₱24,000(e.a)	Central DHCP/DNS for all branches

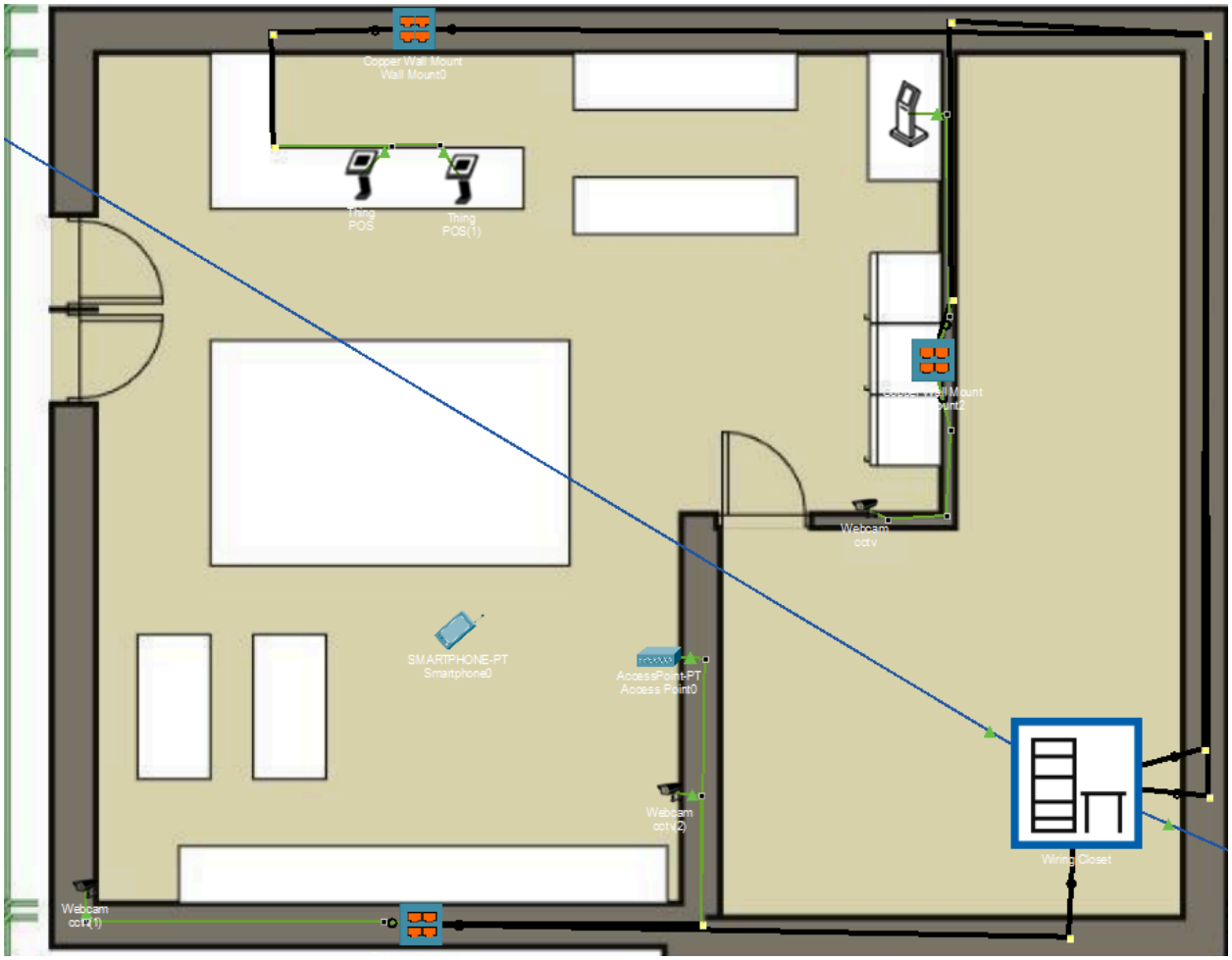
Section 3 Detailed Cabling Floor Plan

Section 3: Detailed Cabling Floor Plan

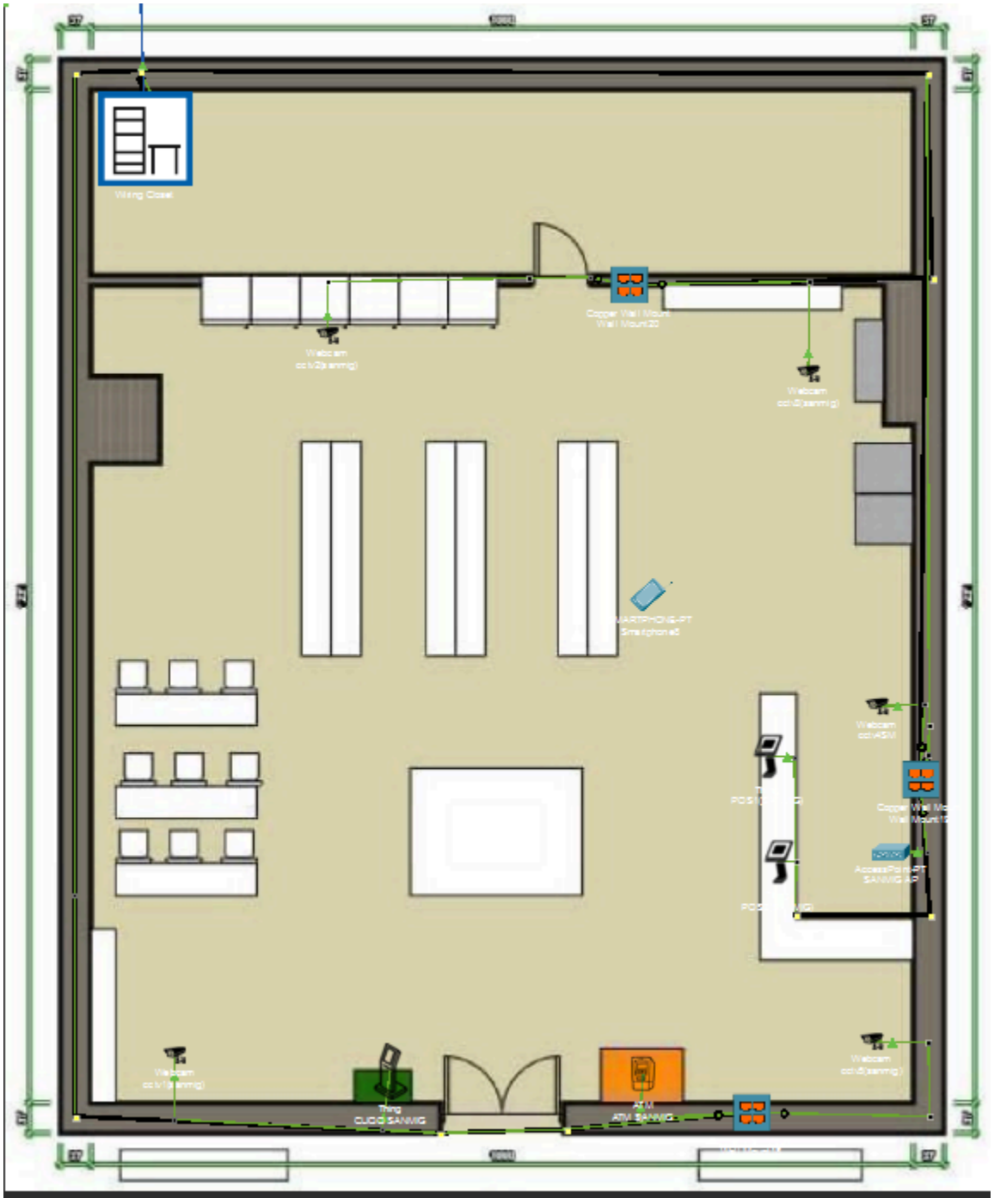
This is a more detailed version of your floor plan. It must show:

- The location of the **Main Distribution Frame (MDF)** / Server Room.
- The location of every **data jack** (wall port).
- A **unique label** for each data jack (e.g., ADM-01, ADM-02, PUB-01).

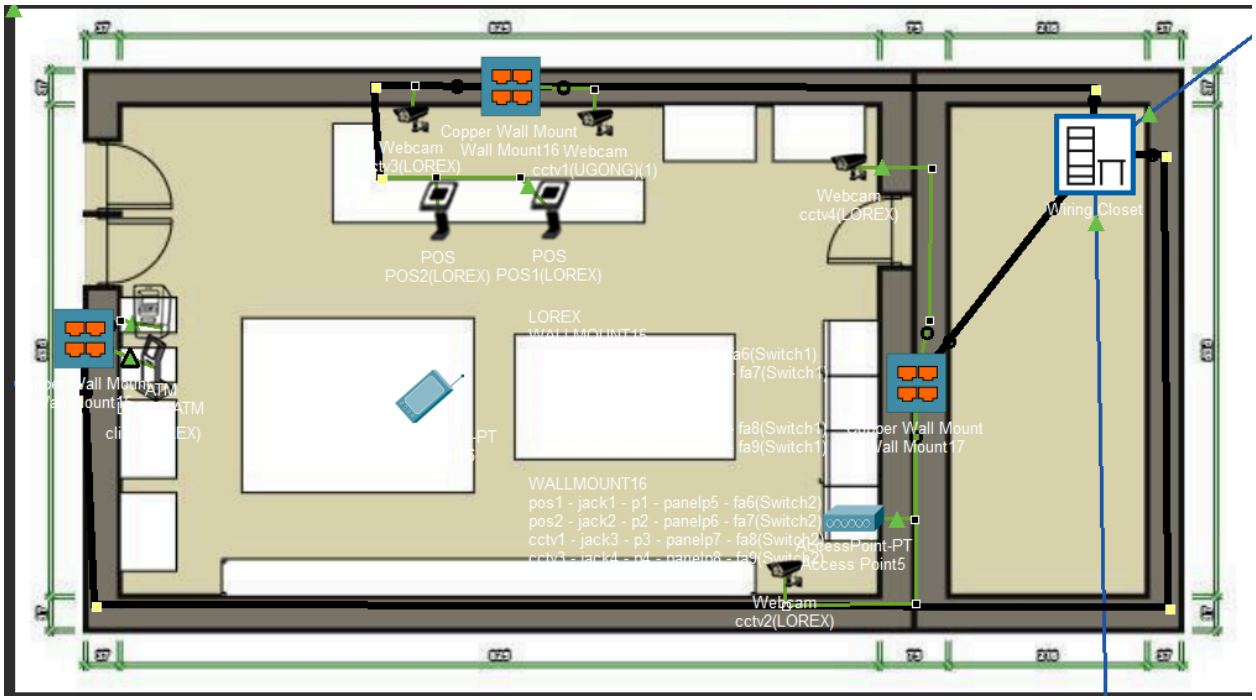
MDF Location



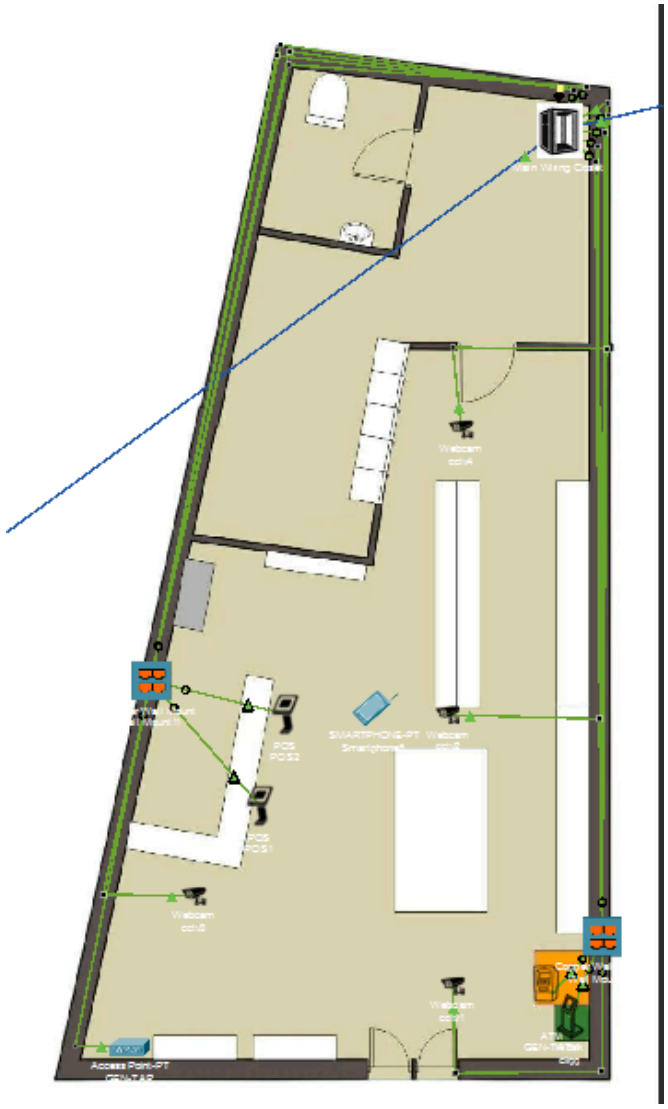
12th Road San Miguel Heights



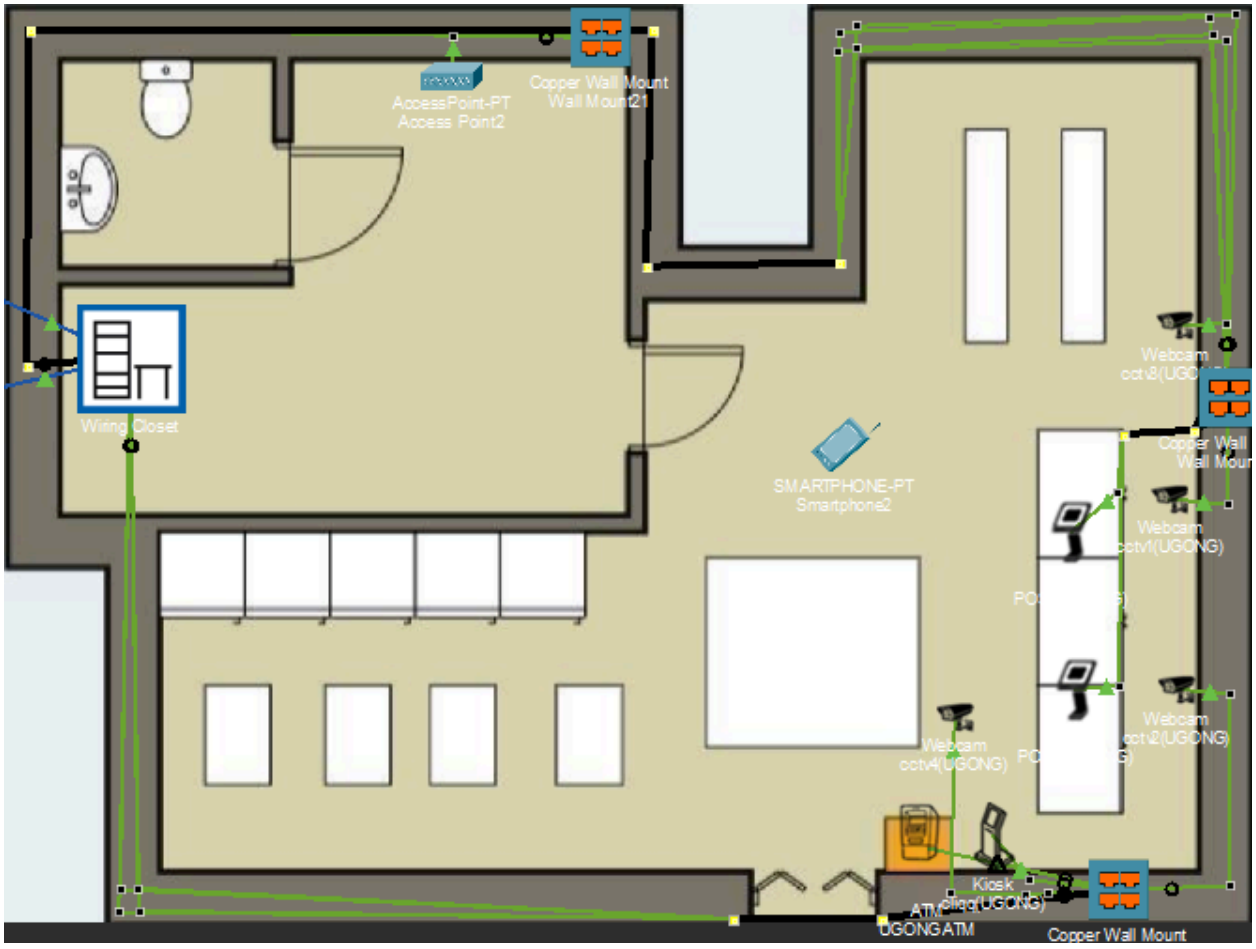
Lorex


Copper Wall Mount
Wall Mount16
Webcam
cctv1(UGONG)(1)
POS
POS2(LOREX)
POS
POS1(LOREX)
Webcam
cctv4(LOREX)
Wiring Closet
LOREX
WALLMOUNT16
pos1 - jack1 - p1 - panelp5 - fa6(Switch2)
pos2 - jack2 - p2 - panelp6 - fa7(Switch2)
cctv1 - jack3 - p3 - panelp7 - fa8(Switch2)
Access Point5
Webcam
cctv2(LOREX)
Wall Mount17

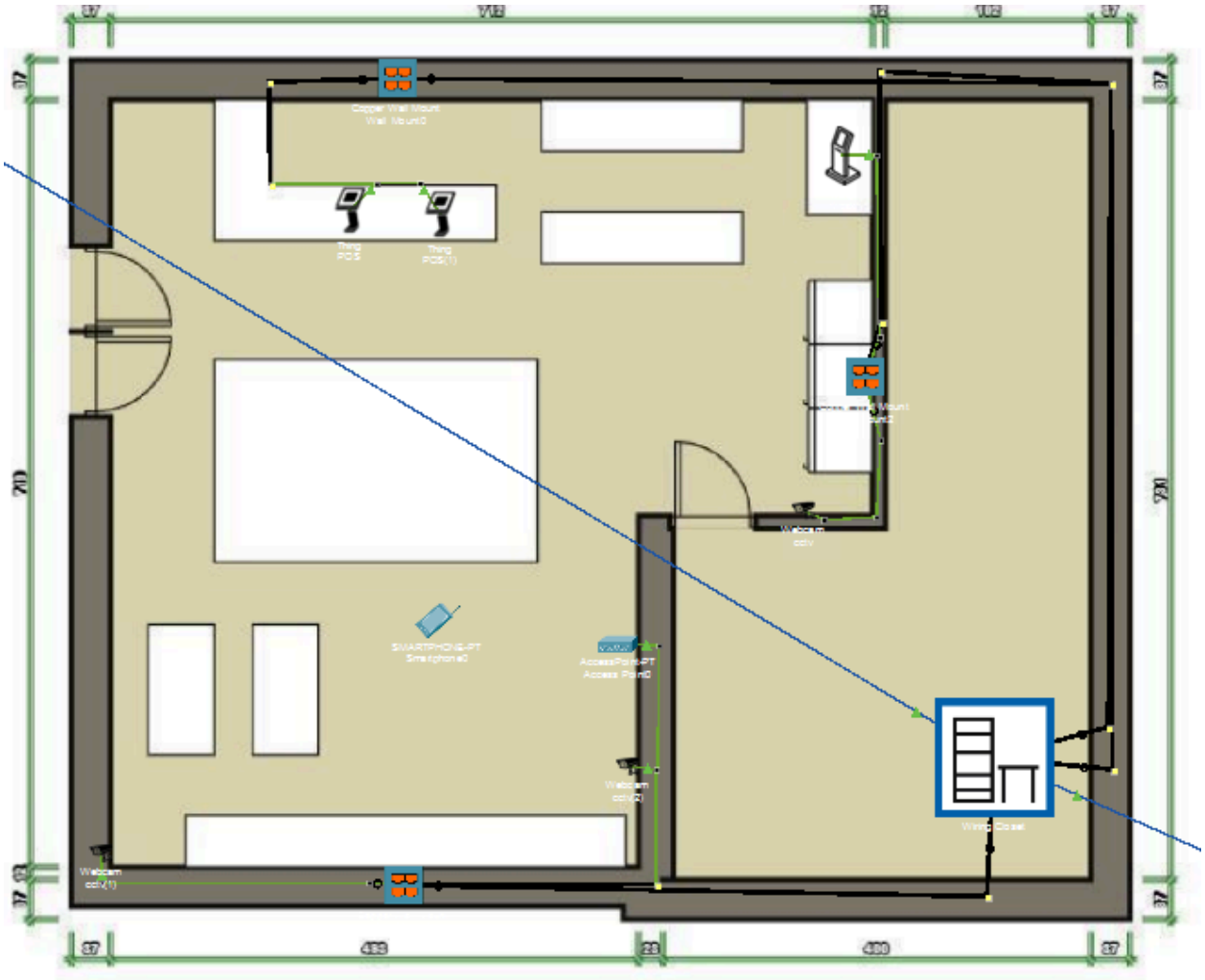
Gen T.



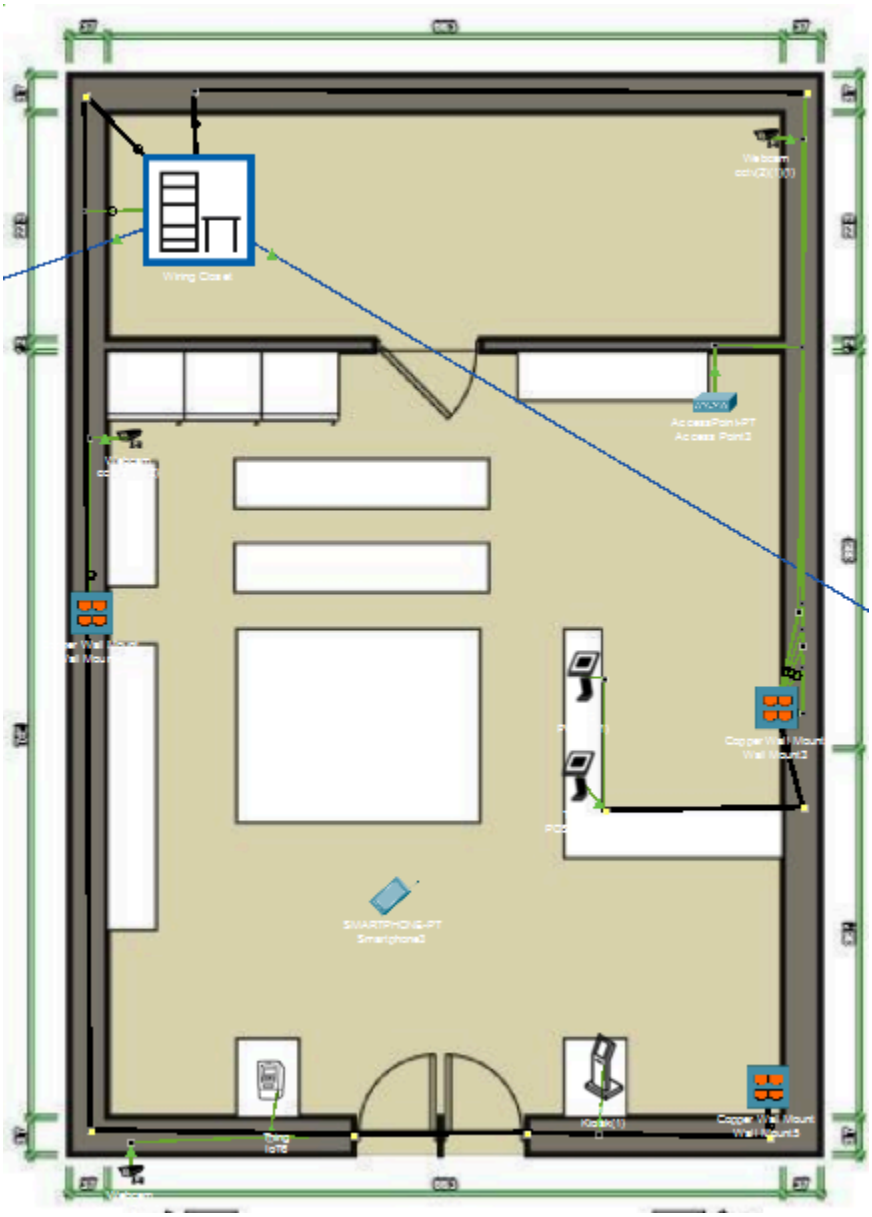
Ugong


AccessPoint-PT
Access Point2
Copper Wall Mount
Wall Mount21
Wiring Closet
SMARTPHONE-PT
Smartphone2
Webcam
Webcam
cctv1(UGONG)
Webcam
cctv2(UGONG)
Webcam
cctv4(UGONG)
Kiosk
UGONG ATM
Copper Wall Mount

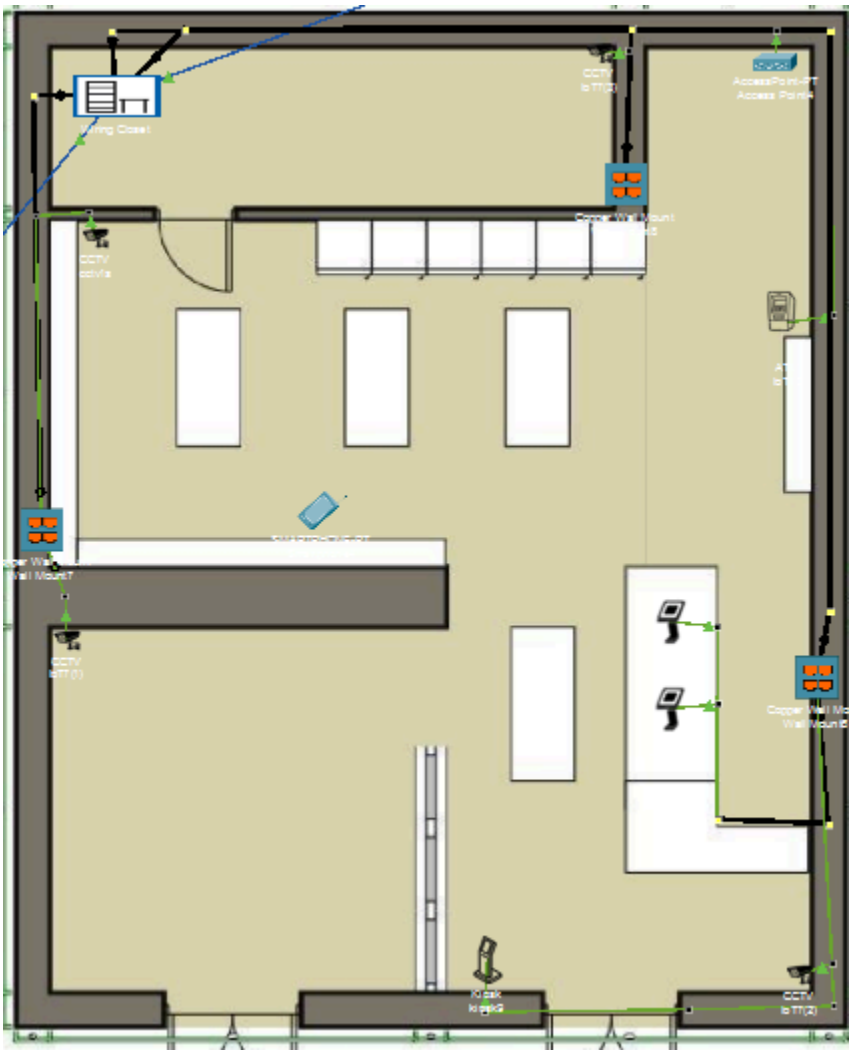
Parada



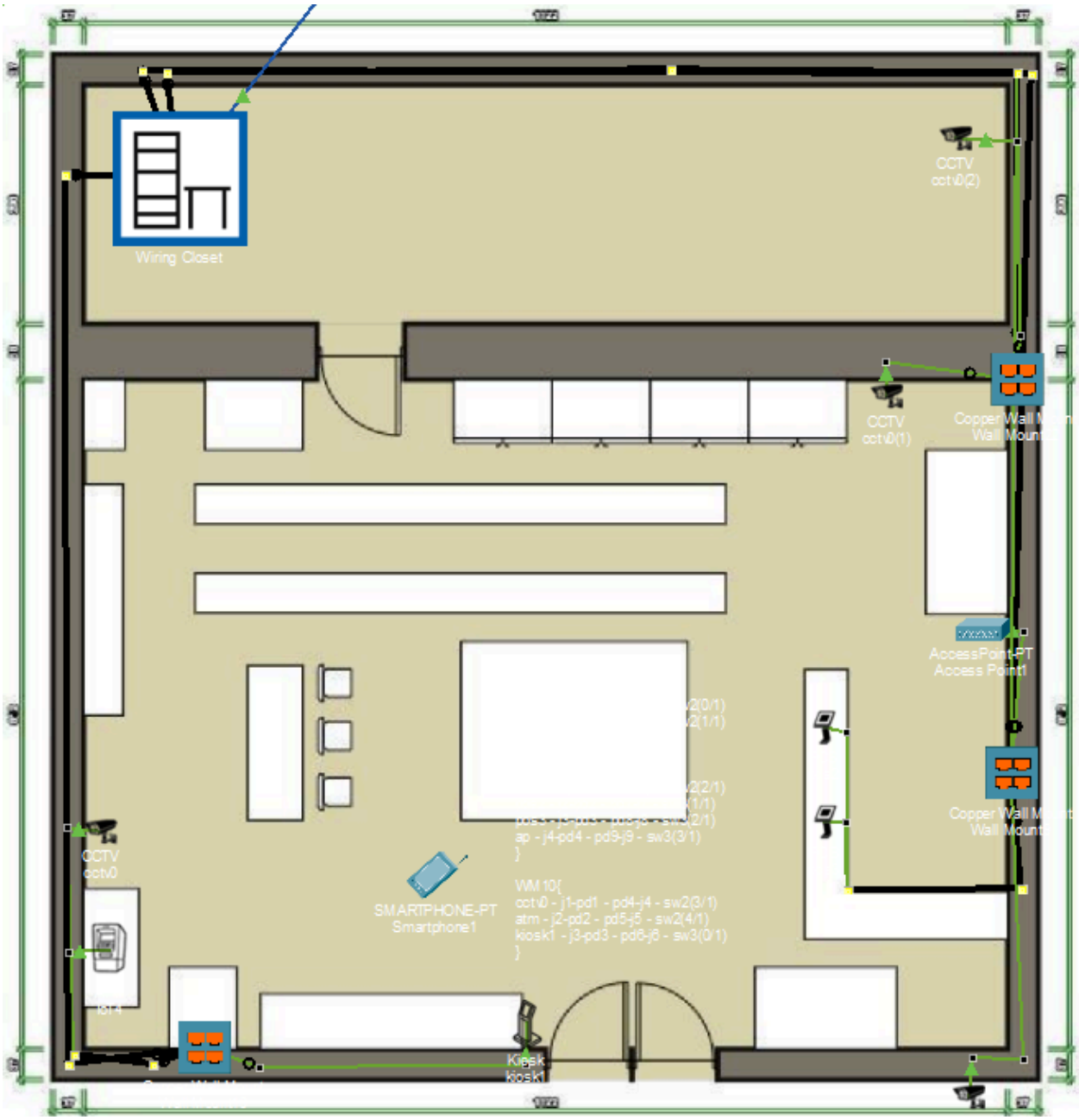
Maysan



Maysan Arca Plaza



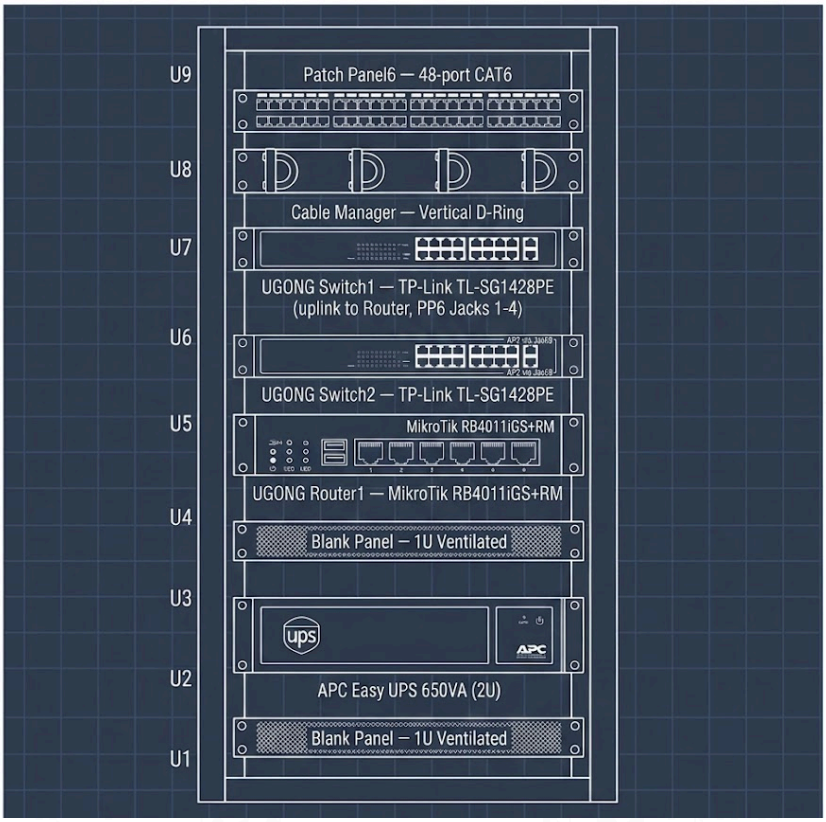
Malinta


Wiring Closet
CCTV
cctv0(2)
CCTV
cctv0(1)
AccessPoint-PT
Access Point1
CCTV
cctv0
SMARTPHONE-PT
Smartphone1
ap - j4-pd4 - pd9-j9 - sw3(3/1)
}
VM10{
cctv0 - j1-pd1 - pd4-j4 - sw2(3/1)
atm - j2-pd2 - pd5-j5 - sw2(4/1)
kiosk1 - j3-pd3 - pd6-j6 - sw3(0/1)
}
Kiosk
kiosk1

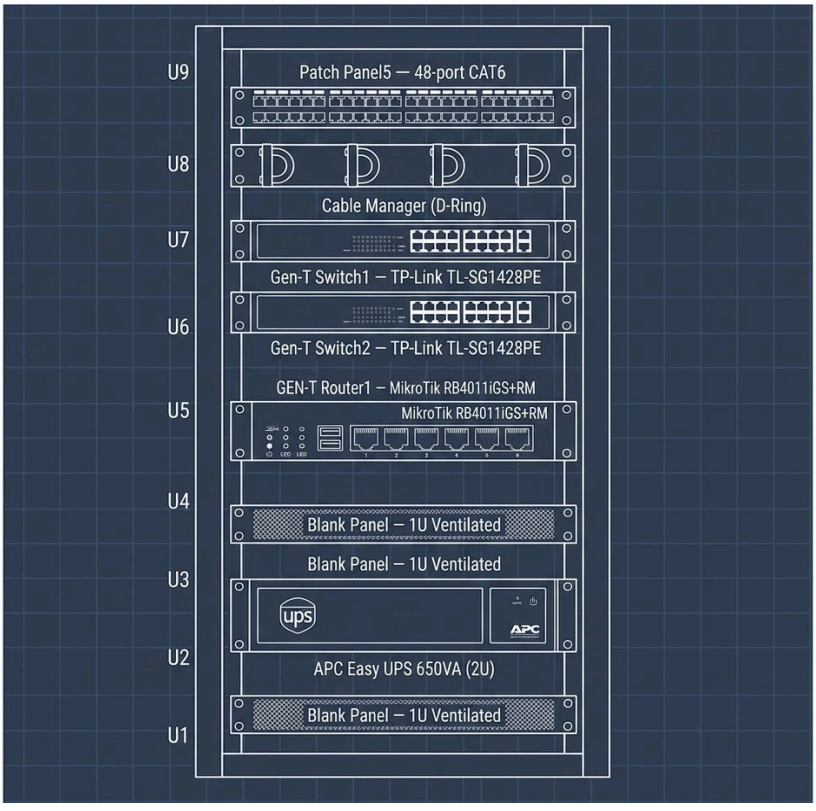
Section 4 Rack Diagram

Section 4: Rack Diagram (MDF Elevation)

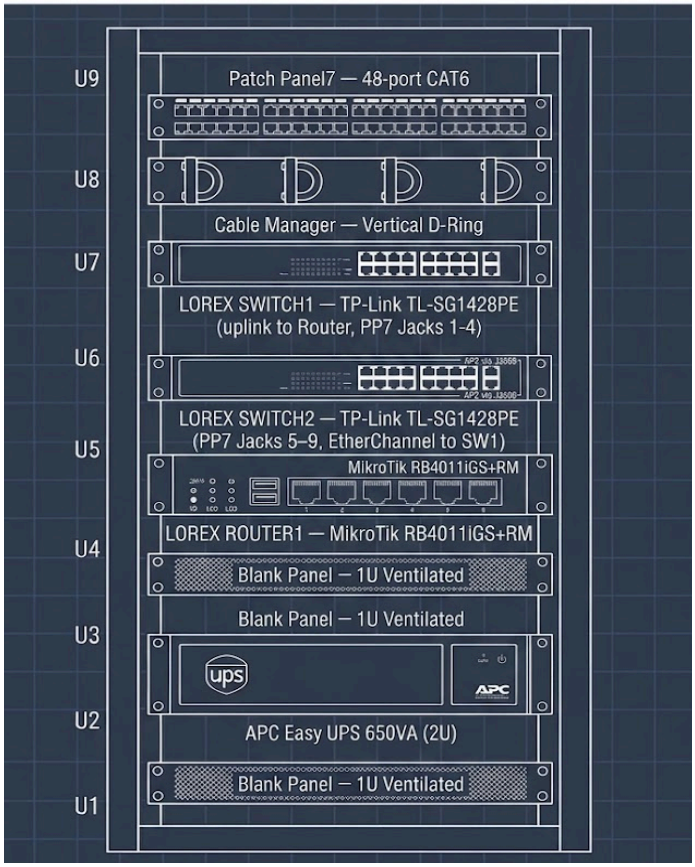
UGONG — MDF Rack Layout (9U)



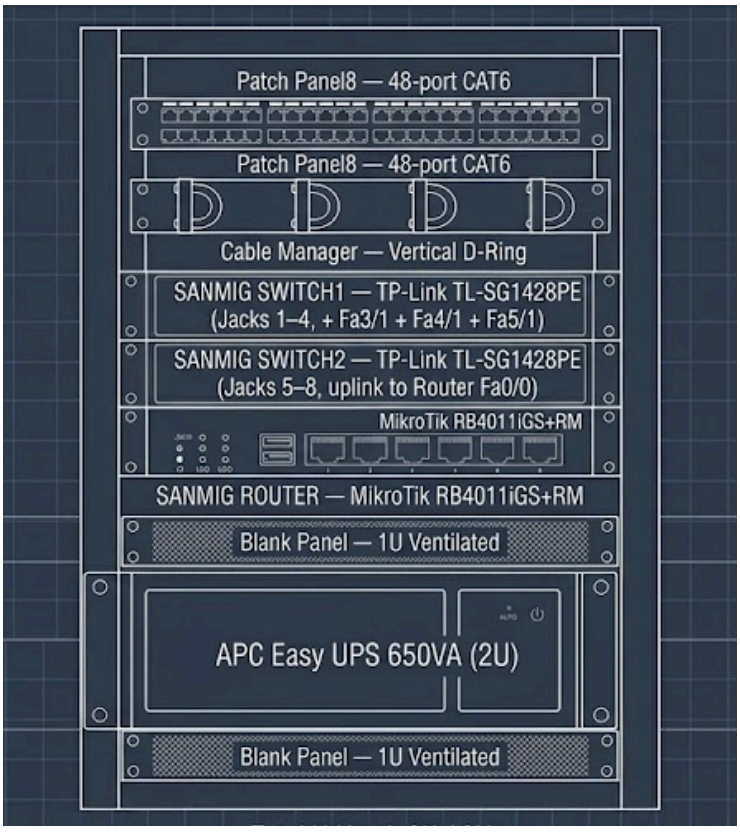
GEN-T — MDF Rack Layout (9U)



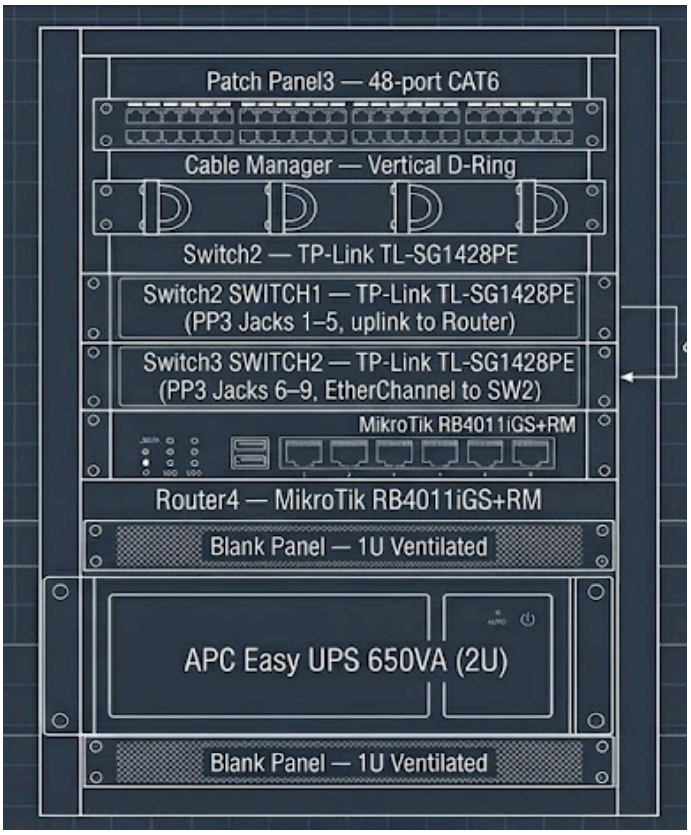
LOREX — MDF Rack Layout (9U)



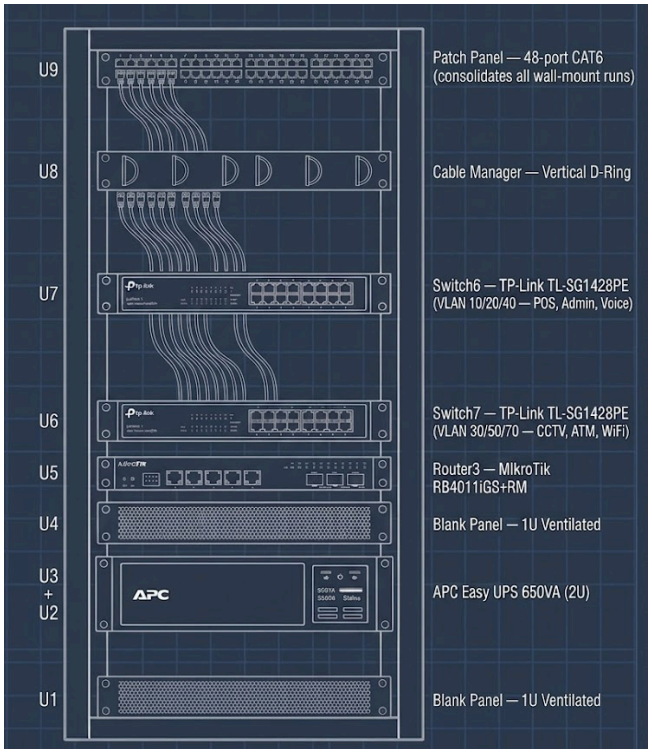
SAN MIGUEL HEIGHTS (SANMIG) — MDF Rack Layout



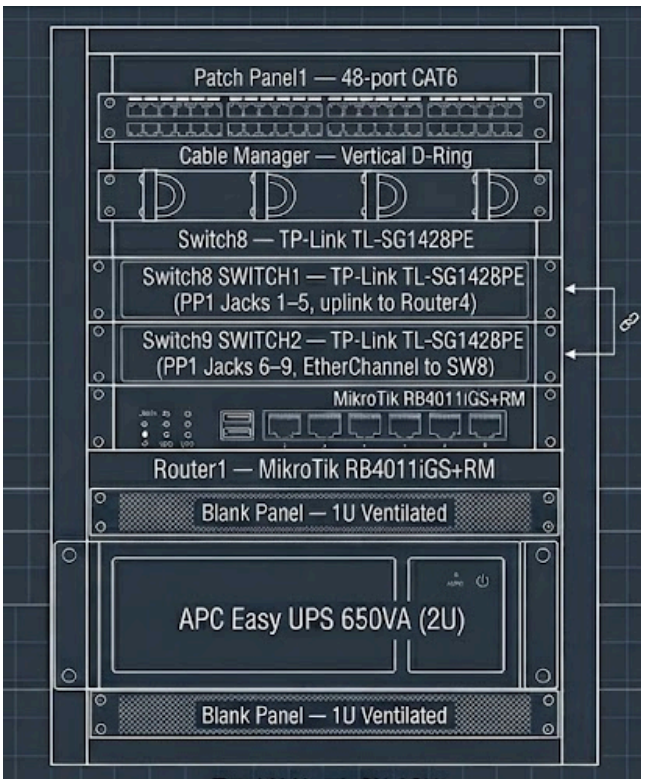
ARCA PLAZA (ARC) — MDF Rack Layout



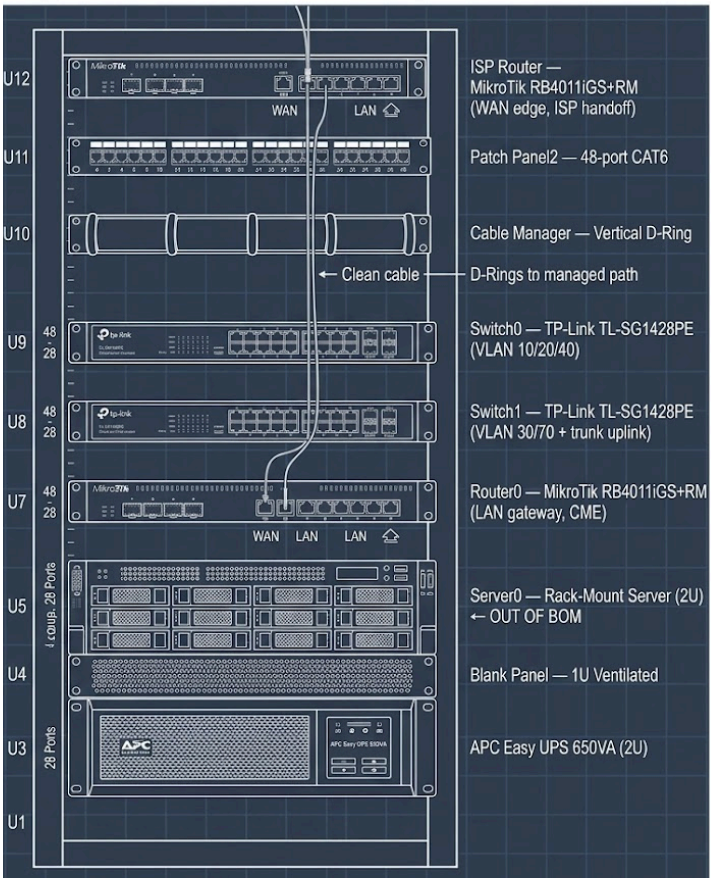
MAYSAN — MDF Rack Layout



MALINTA (MAL) — MDF Rack Layout



PARADA MDF Rack Layout



Section 5 Port Labeling

Section 5: Cable Labeling Scheme & Port Map

Labeling Scheme Rule:
`[BRANCH]-[ZONE]-[JACK#]`

- `PAR` = Parada | `MAY` = Maysan | `GEN` = Gen-T | `UGO` = Ugong | `LOR` = Lorex | `SAN` = San Miguel Heights | `ARC` = Arca Plaza | MAL = Malinta
- `POS` = Point of Sale | `ADM` = Admin/Back Office | `CCTV` = Security Cameras | `VOI` = Voice/Phone | `WIF` = Wi-Fi/AP | `ATM` = ATM Machine | `SRV` = Server/Network Rack

Wall_Jack_ID	Cable_Label	Wall_Mount	Jack_No	Patch_Panel_Port	Switch_Name	Switch_Port	VLAN_ID	VLAN_Name	Device_Name	Device_Type	Device_Purpose
PAR-SRV-01	PAR-SRV-01	DIRECT	N/A	DIRECT	Switch0	Fa0/1	1	Management	Server0	Server	Server/Network Rack
PAR-POS-01	PAR-POS-01	WM0	J1	PP2-J5	Switch0	Fa1/1	10	POS	POS	POS Terminal	Point of Sale
PAR-POS-02	PAR-POS-02	WM0	J2	PP2-J6	Switch0	Fa2/1	10	POS	POS(1)	POS Terminal	Point of Sale
PAR-POS-03	PAR-POS-03	WM2	J2	PP2-J7	Switch0	Fa3/1	10	POS	Kiosk	Kiosk	Point of Sale

PAR-VOI-01	PAR-VOI-01	DIRECT	N/A	DIRECT	Switch0	Fa4/1	40	Voice	IP Phone0	IP Phone	Voice/Phone
PAR-VOI-01(PC)	PAR-VOI-01(PC)	DIRECT	N/A	PASSTHROUGH	Switch0	Fa4/1	20	Admin	PC0	PC	Admin/Back Office
PAR-SRV-02	PAR-SRV-02	DIRECT	N/A	DIRECT	Switch0	Fa5/1	N/A	Trunk	Switch1	Switch	Inter-switch EtherChannel
PAR-SRV-03	PAR-SRV-03	DIRECT	N/A	DIRECT	Switch0	Fa6/1	N/A	Trunk	Switch1	Switch	Inter-switch EtherChannel
PAR-CCTV-01	PAR-CCTV-01	WM2	J1	PP2-J1	Switch1	Fa0/1	30	CCTV	cctv	IP Camera	Security Camera
PAR-CCTV-02	PAR-CCTV-02	WM1	J1	PP2-J2	Switch1	Fa1/1	30	CCTV	cctv(1)	IP Camera	Security Camera
PAR-CCTV-03	PAR-CCTV-03	WM1	J2	PP2-J3	Switch1	Fa2/1	30	CCTV	cctv(2)	IP Camera	Security Camera
PAR-WIF-01	PAR-WIF-01	WM1	J3	PP2-J4	Switch1	Fa3/1	70	Wi-Fi	Access Point0	Access Point	Wi-Fi/AP

PAR-SRV-04	PAR-SRV-04	DIRECT	N/A	DIRECT	Switch1	Fa4/1	N/A	Trunk	Router0	Router	Router Uplink
PAR-SRV-05	PAR-SRV-05	DIRECT	N/A	DIRECT	Switch1	Fa5/1	N/A	Trunk	Switch0	Switch	Inter-switch EtherChannel
PAR-SRV-06	PAR-SRV-06	DIRECT	N/A	DIRECT	Switch1	Fa6/1	N/A	Trunk	Switch0	Switch	Inter-switch EtherChannel
MAY-CCTV-01	MAY-CCTV-01	WM4	J1	PP0-J1	Switch7	Fa0/1	30	CCTV	cctv(2)(1)(2)	IP Camera	Security Camera
MAY-CCTV-02	MAY-CCTV-02	WM3	J1	PP0-J2	Switch7	Fa1/1	30	CCTV	cctv(2)(1)(1)	IP Camera	Security Camera
MAY-CCTV-03	MAY-CCTV-03	WM5	J1	PP0-J3	Switch7	Fa2/1	30	CCTV	cctv(2)(1)	IP Camera	Security Camera
MAY-ATM-01	MAY-ATM-01	WM5	J2	PP0-J4	Switch7	Fa3/1	50	ATM	IoT6	ATM	ATM Machine
MAY-WIF-01	MAY-WIF-01	WM3	J2	PP0-J5	Switch7	Fa4/1	70	Wi-Fi	Access Point3	Access Point	Wi-Fi/AP

MAY-SRV-01	MAY-SRV-01	DIRECT	N/A	DIRECT	Switch7	Fa5/1	N/A	Trunk	Switch6	Switch	Inter-switch EtherChannel
MAY-SRV-02	MAY-SRV-02	DIRECT	N/A	DIRECT	Switch7	Fa9/1	N/A	Trunk	Switch6	Switch	Inter-switch EtherChannel
MAY-POS-01	MAY-POS-01	WM3	J3	PP0-J6	Switch6	Fa0/1	10	POS	POS(1)(1)	POS Terminal	Point of Sale
MAY-POS-02	MAY-POS-02	WM3	J4	PP0-J7	Switch6	Fa1/1	10	POS	POS(1)(1)(1)	POS Terminal	Point of Sale
MAY-POS-03	MAY-POS-03	WM5	J3	PP0-J8	Switch6	Fa2/1	10	POS	Kiosk(1)	Kiosk	Point of Sale
MAY-VOI-01	MAY-VOI-01	DIRECT	N/A	DIRECT	Switch6	Fa3/1	40	Voice	IP Phone3	IP Phone	Voice/Phone
MAY-VOI-01(PC)	MAY-VOI-01(PC)	DIRECT	N/A	PASSTHROUGH	Switch6	Fa3/1	20	Admin	PC3	PC	Admin/Back Office
MAY-SRV-03	MAY-SRV-03	DIRECT	N/A	DIRECT	Switch6	Fa7/1	N/A	Trunk	Router3	Router	Router Uplink

MAY-SRV-04	MAY-SRV-04	DIRECT	N/A	DIRECT	Switch6	Fa8/1	N/A	Trunk	Switch7	Switch	Inter-switch EtherChannel
MAY-SRV-05	MAY-SRV-05	DIRECT	N/A	DIRECT	Switch6	Fa9/1	N/A	Trunk	Switch7	Switch	Inter-switch EtherChannel
GEN-POS-01	GEN-POS-01	WM11	J1	PP5-J1	Gen-T Switch1	Fa6/1	10	POS	POS2	POS Terminal	Point of Sale
GEN-POS-02	GEN-POS-02	WM11	J2	PP5-J2	Gen-T Switch1	Fa7/1	10	POS	POS1	POS Terminal	Point of Sale
GEN-CCTV-01	GEN-CCTV-01	WM11	J4	PP5-J3	Gen-T Switch1	Fa8/1	30	CCTV	cctv3	IP Camera	Security Camera
GEN-WIF-01	GEN-WIF-01	WM11	J3	PP5-J4	Gen-T Switch1	Fa9/1	70	Wi-Fi	GEN-T AP	Access Point	Wi-Fi/AP
GEN-VOI-01	GEN-VOI-01	DIRECT	N/A	DIRECT	Gen-T Switch1	Fa3/1	40	Voice	GEN-T Phone	IP Phone	Voice/Phone
GEN-VOI-01(PC)	GEN-VOI-01(PC)	DIRECT	N/A	PASSTHROUGH	Gen-T Switch1	Fa3/1	20	Admin	GEN-T PC1	PC	Admin/Back Office

GEN-ATM-01	GEN-ATM-01	WM12	J1	PP5-J5	Gen-T Switch2	Fa6/1	50	ATM	GEN-T ATM	ATM	ATM Machine
GEN-POS-03	GEN-POS-03	WM12	J2	PP5-J6	Gen-T Switch2	Fa7/1	10	POS	cliqq	Kiosk	Point of Sale
GEN-CCTV-02	GEN-CCTV-02	WM12	J3	PP5-J7	Gen-T Switch2	Fa8/1	30	CCTV	cctv1	IP Camera	Security Camera
GEN-CCTV-03	GEN-CCTV-03	WM12	J4	PP5-J8	Gen-T Switch2	Fa9/1	30	CCTV	cctv2	IP Camera	Security Camera
GEN-CCTV-04	GEN-CCTV-04	DIRECT	N/A	DIRECT	Gen-T Switch2	Fa0/1	30	CCTV	cctv4	IP Camera	Security Camera
GEN-SRV-01	GEN-SRV-01	DIRECT	N/A	DIRECT	Gen-T Switch1	Fa0/1	N/A	Trunk	GEN-T Router1	Router	Router Uplink
GEN-SRV-02	GEN-SRV-02	DIRECT	N/A	DIRECT	Gen-T Switch1	Fa1/1	N/A	Trunk	Gen-T Switch2	Switch	Inter-switch Trunk
GEN-SRV-03	GEN-SRV-03	DIRECT	N/A	DIRECT	Gen-T Switch1	Fa2/1	N/A	Trunk	Gen-T Switch2	Switch	Inter-switch Trunk

UGO-ATM-01	UGO-ATM-01	WM13	J1	PP6-J1	UGONG Switch1	Fa6/1	50	ATM	UGONG ATM	ATM	ATM Machine
UGO-POS-01	UGO-POS-01	WM13	J2	PP6-J2	UGONG Switch1	Fa7/1	10	POS	cliqq(UGONG)	Kiosk	Point of Sale
UGO-CCTV-01	UGO-CCTV-01	WM13	J3	PP6-J3	UGONG Switch1	Fa8/1	30	CCTV	cctv2(UGONG)	IP Camera	Security Camera
UGO-CCTV-02	UGO-CCTV-02	WM13	J4	PP6-J4	UGONG Switch1	Fa9/1	30	CCTV	cctv4(UGONG)	IP Camera	Security Camera
UGO-SRV-01	UGO-SRV-01	DIRECT	N/A	DIRECT	UGONG Switch1	Fa0/1	N/A	Trunk	UGONG Router1	Router	Router Uplink
UGO-POS-02	UGO-POS-02	WM14	J1	PP6-J5	UGONG Switch2	Fa6/1	10	POS	POS2(UGONG)	POS Terminal	Point of Sale
UGO-POS-03	UGO-POS-03	WM14	J2	PP6-J6	UGONG Switch2	Fa7/1	10	POS	POS1(UGONG)	POS Terminal	Point of Sale
UGO-CCTV-03	UGO-CCTV-03	WM14	J3	PP6-J7	UGONG Switch2	Fa8/1	30	CCTV	cctv1(UGONG)	IP Camera	Security Camera

UGO-CCTV-04	UGO-CCTV-04	WM14	J4	PP6-J8	UGONG Switch2	Fa9/1	30	CCTV	cctv3(UGONG)	IP Camera	Security Camera
UGO-WIF-01	UGO-WIF-01	WM21	J1	PP6-J9	UGONG Switch2	Fa3/1	70	Wi-Fi	Access Point2	Access Point	Wi-Fi/AP
UGO-VOI-01	UGO-VOI-01	DIRECT	N/A	DIRECT	UGONG Switch2	Fa0/1	40	Voice	UGONG PHONE	IP Phone	Voice/Phone
UGO-VOI-01(PC)	UGO-VOI-01 (PC)	DIRECT	N/A	PASSTHROUGH	UGONG Switch2	Fa0/1	20	Admin	UGONGPC	PC	Admin/Back Office
UGO-VOI-02	UGO-VOI-02	DIRECT	N/A	DIRECT	LOREX SWITCH2	Fa0/1	40	Voice	UGONG PHONE(1)	IP Phone	Voice/Phone
UGO-VOI-02(PC)	UGO-VOI-02 (PC)	DIRECT	N/A	PASSTHROUGH	LOREX SWITCH2	Fa0/1	20	Admin	UGONGPC(1)	PC	Admin/Back Office
LOR-ATM-01	LOR-ATM-01	WM15	J1	PP7-J1	LOREX SWITCH1	Fa6/1	50	ATM	LOREX ATM	ATM	ATM Machine
LOR-POS-01	LOR-POS-01	WM15	J2	PP7-J2	LOREX SWITCH1	Fa7/1	10	POS	cliqq(LOREX)	Kiosk	Point of Sale

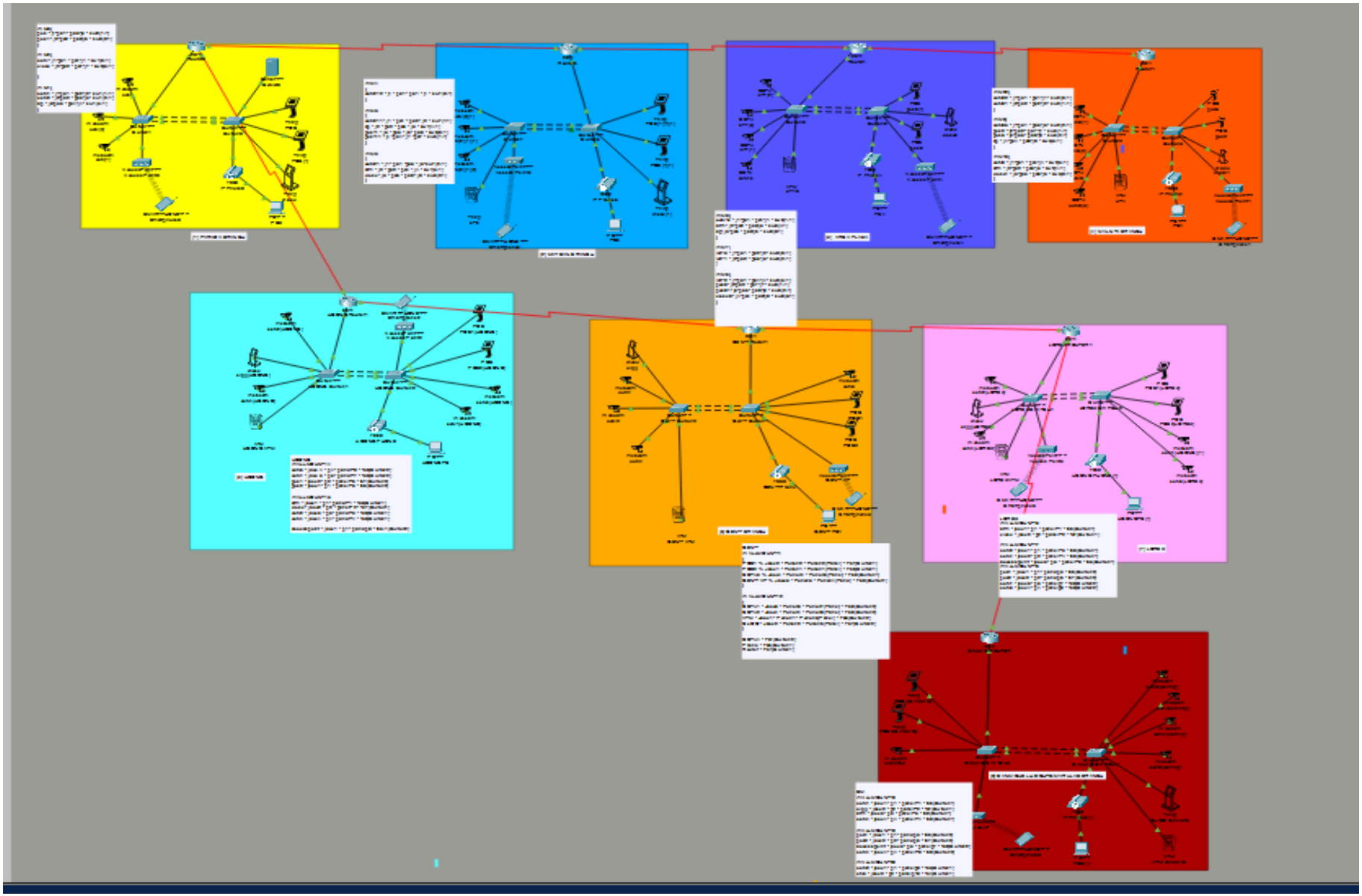
LOR-CCTV-01	LOR-CCTV-01	WM17	J1	PP7-J3	LOREX SWITCH1	Fa8/1	30	CCTV	cctv2(LOREX)	IP Camera	Security Camera
LOR-CCTV-02	LOR-CCTV-02	WM17	J2	PP7-J4	LOREX SWITCH1	Fa9/1	30	CCTV	cctv4(LOREX)	IP Camera	Security Camera
LOR-WIF-01	LOR-WIF-01	WM17	J3	PP7-J9	LOREX SWITCH1	Fa3/1	70	Wi-Fi	Access Point5	Access Point	Wi-Fi/AP
LOR-SRV-01	LOR-SRV-01	DIRECT	N/A	DIRECT	LOREX SWITCH1	Fa0/1	N/A	Trunk	LOREX ROUTER1	Router	Router Uplink
LOR-POS-02	LOR-POS-02	WM16	J1	PP7-J5	LOREX SWITCH2	Fa6/1	10	POS	POS1(LOREX)	POS Terminal	Point of Sale
LOR-POS-03	LOR-POS-03	WM16	J2	PP7-J6	LOREX SWITCH2	Fa7/1	10	POS	POS2(LOREX)	POS Terminal	Point of Sale
LOR-CCTV-03	LOR-CCTV-03	WM16	J3	PP7-J7	LOREX SWITCH2	Fa8/1	30	CCTV	cctv1(UGONG)(1)	IP Camera	Security Camera
LOR-CCTV-04	LOR-CCTV-04	WM16	J4	PP7-J8	LOREX SWITCH2	Fa9/1	30	CCTV	cctv3(LOREX)	IP Camera	Security Camera

SAN-CCTV-01	SAN-CCTV-01	WM18	J1	PP8-J1	SANMIG SWITCH1	Fa6/1	30	CCTV	cctv1(sanmig)	IP Camera	Security Camera
SAN-POS-01	SAN-POS-01	WM18	J2	PP8-J2	SANMIG SWITCH1	Fa7/1	10	POS	CLIQQ SANMIG	Kiosk	Point of Sale
SAN-ATM-01	SAN-ATM-01	WM18	J3	PP8-J3	SANMIG SWITCH1	Fa8/1	50	ATM	ATM SANMIG	ATM	ATM Machine
SAN-CCTV-02	SAN-CCTV-02	WM18	J4	PP8-J4	SANMIG SWITCH1	Fa9/1	30	CCTV	cctv5(sanmig)	IP Camera	Security Camera
SAN-CCTV-03	SAN-CCTV-03	WM20	J1	PP8-J9	SANMIG SWITCH1	Fa3/1	30	CCTV	cctv2(sanmig)	IP Camera	Security Camera
SAN-CCTV-04	SAN-CCTV-04	WM20	J2	PP8-J10	SANMIG SWITCH1	Fa4/1	30	CCTV	cctv3(sanmig)	IP Camera	Security Camera
SAN-VOI-01	SAN-VOI-01	DIRECT	N/A	DIRECT	SANMIG SWITCH1	Fa5/1	40	Voice	IP Phone0(1)	IP Phone	Voice/Phone
SAN-VOI-01(PC)	SAN-VOI-01(PC)	DIRECT	N/A	PASSTHROUGH	SANMIG SWITCH1	Fa5/1	20	Admin	PC0(1)	PC	Admin/Back Office

SAN-POS-02	SAN-POS-02	WM19	J1	PP8-J5	SANMIG SWITCH2	Fa6/1	10	POS	POS1(SANMIG)	POS Terminal	Point of Sale
SAN-POS-03	SAN-POS-03	WM19	J2	PP8-J6	SANMIG SWITCH2	Fa7/1	10	POS	POS2(SANMIG)	POS Terminal	Point of Sale
SAN-WIF-01	SAN-WIF-01	WM19	J3	PP8-J7	SANMIG SWITCH2	Fa8/1	70	Wi-Fi	SANMIG AP	Access Point	Wi-Fi/AP
SAN-CCTV-05	SAN-CCTV-05	WM19	J4	PP8-J8	SANMIG SWITCH2	Fa9/1	30	CCTV	cctv4SM	IP Camera	Security Camera
SAN-SRV-01	SAN-SRV-01	DIRECT	N/A	DIRECT	SANMIG SWITCH2	Fa0/1	N/A	Trunk	SANMIG ROUTER	Router	Router Uplink

Section 6 Network Topology

Section 6: Network Topology



Section 7 REAL

📍 Branch 1 — PARADA (10.100.1.0/24)

VLAN	Purpose	Net. Address	Prefix	Subnet Mask	Hosts Needed	First Usable	Last Usable	Broadcast	Suggested Gateway
VLAN 1	Static IP (Router/Switch/Server)	10.100.1.0	/28	255.255.255.240	11	10.100.1.1	10.100.1.14	10.100.1.15	10.100.1.1
VLAN 30	CCTV	10.100.1.16	/28	255.255.255.240	11	10.100.1.17	10.100.1.30	10.100.1.31	10.100.1.17
VLAN 70	WiFi	10.100.1.32	/28	255.255.255.240	11	10.100.1.33	10.100.1.46	10.100.1.47	10.100.1.33
VLAN 10	POS	10.100.1.48	/29	255.255.255.248	4	10.100.1.49	10.100.1.54	10.100.1.55	10.100.1.49
VLAN 20	PC Admin	10.100.1.56	/29	255.255.255.248	4	10.100.1.57	10.100.1.62	10.100.1.63	10.100.1.57
VLAN 40	VoIP	10.100.1.64	/29	255.255.255.248	4	10.100.1.65	10.100.1.70	10.100.1.71	10.100.1.65
VLAN 50	ATM	10.100.1.72	/29	255.255.255.248	4	10.100.1.73	10.100.1.78	10.100.1.79	10.100.1.73
P2P-1	Branch 1 ↔ Branch 2	10.100.1.80	/30	255.255.255.252	2	10.100.1.81	10.100.1.82	10.100.1.83	10.100.1.81
P2P-2	Branch ↔ ISP/Core	10.100.1.84	/30	255.255.255.252	2	10.100.1.85	10.100.1.86	10.100.1.87	10.100.1.85

📍 Branch 2 — MAYSAN (10.100.2.0/24)

VLAN	Purpose	Net. Address	Prefix	Subnet Mask	Hosts Needed	First Usable	Last Usable	Broadcast	Suggested Gateway
VLAN 1	Static IP (Router/Switch/Server)	10.100.2.0	/28	255.255.255.240	11	10.100.2.1	10.100.2.14	10.100.2.15	10.100.2.1
VLAN 30	CCTV	10.100.2.16	/28	255.255.255.240	11	10.100.2.17	10.100.2.30	10.100.2.31	10.100.2.17
VLAN 70	WiFi	10.100.2.32	/28	255.255.255.240	11	10.100.2.33	10.100.2.46	10.100.2.47	10.100.2.33
VLAN 10	POS	10.100.2.48	/29	255.255.255.248	4	10.100.2.49	10.100.2.54	10.100.2.55	10.100.2.49
VLAN 20	PC Admin	10.100.2.56	/29	255.255.255.248	4	10.100.2.57	10.100.2.62	10.100.2.63	10.100.2.57
VLAN 40	VoIP	10.100.2.64	/29	255.255.255.248	4	10.100.2.65	10.100.2.70	10.100.2.71	10.100.2.65
VLAN 50	ATM	10.100.2.72	/29	255.255.255.248	4	10.100.2.73	10.100.2.78	10.100.2.79	10.100.2.73

📍 Branch 3 — GEN-T (10.100.3.0/24)

VLAN	Purpose	Net. Address	Prefix	Subnet Mask	Hosts Needed	First Usable	Last Usable	Broadcast	Suggested Gateway
VLAN 1	Static IP (Router/Switch/Server)	10.100.3.0	/28	255.255.255.240	11	10.100.3.1	10.100.3.14	10.100.3.15	10.100.3.1
VLAN 30	CCTV	10.100.3.16	/28	255.255.255.240	11	10.100.3.17	10.100.3.30	10.100.3.31	10.100.3.17
VLAN 70	WiFi	10.100.3.32	/28	255.255.255.240	11	10.100.3.33	10.100.3.46	10.100.3.47	10.100.3.33
VLAN 10	POS	10.100.3.48	/29	255.255.255.248	4	10.100.3.49	10.100.3.54	10.100.3.55	10.100.3.49
VLAN 20	PC Admin	10.100.3.56	/29	255.255.255.248	4	10.100.3.57	10.100.3.62	10.100.3.63	10.100.3.57
VLAN 40	VoIP	10.100.3.64	/29	255.255.255.248	4	10.100.3.65	10.100.3.70	10.100.3.71	10.100.3.65
VLAN 50	ATM	10.100.3.72	/29	255.255.255.248	4	10.100.3.73	10.100.3.78	10.100.3.79	10.100.3.73

📍 Branch 4 — UGONG (10.100.4.0/24)

VLAN	Purpose	Net. Address	Prefix	Subnet Mask	Hosts Needed	First Usable	Last Usable	Broadcast	Suggested Gateway
VLAN 1	Static IP (Router/Switch/Server)	10.100.4.0	/28	255.255.255.240	11	10.100.4.1	10.100.4.14	10.100.4.15	10.100.4.1
VLAN 30	CCTV	10.100.4.16	/28	255.255.255.240	11	10.100.4.17	10.100.4.30	10.100.4.31	10.100.4.17
VLAN 70	WiFi	10.100.4.32	/28	255.255.255.240	11	10.100.4.33	10.100.4.46	10.100.4.47	10.100.4.33
VLAN 10	POS	10.100.4.48	/29	255.255.255.248	4	10.100.4.49	10.100.4.54	10.100.4.55	10.100.4.49
VLAN 20	PC Admin	10.100.4.56	/29	255.255.255.248	4	10.100.4.57	10.100.4.62	10.100.4.63	10.100.4.57
VLAN 40	VoIP	10.100.4.64	/29	255.255.255.248	4	10.100.4.65	10.100.4.70	10.100.4.71	10.100.4.65
VLAN 50	ATM	10.100.4.72	/29	255.255.255.248	4	10.100.4.73	10.100.4.78	10.100.4.79	10.100.4.73

📍 Branch 5 — LOREX (10.100.5.0/24)

VLAN	Purpose	Net. Address	Prefix	Subnet Mask	Hosts Needed	First Usable	Last Usable	Broadcast	Suggested Gateway
VLAN 1	Static IP (Router/Switch/Server)	10.100.5.0	/28	255.255.255.240	11	10.100.5.1	10.100.5.14	10.100.5.15	10.100.5.1
VLAN 30	CCTV	10.100.5.16	/28	255.255.255.240	11	10.100.5.17	10.100.5.30	10.100.5.31	10.100.5.17
VLAN 70	WiFi	10.100.5.32	/28	255.255.255.240	11	10.100.5.33	10.100.5.46	10.100.5.47	10.100.5.33
VLAN 10	POS	10.100.5.48	/29	255.255.255.248	4	10.100.5.49	10.100.5.54	10.100.5.55	10.100.5.49
VLAN 20	PC Admin	10.100.5.56	/29	255.255.255.248	4	10.100.5.57	10.100.5.62	10.100.5.63	10.100.5.57
VLAN 40	VoIP	10.100.5.64	/29	255.255.255.248	4	10.100.5.65	10.100.5.70	10.100.5.71	10.100.5.65
VLAN 50	ATM	10.100.5.72	/29	255.255.255.248	4	10.100.5.73	10.100.5.78	10.100.5.79	10.100.5.73

📍 Branch 6 — SAN MIGUEL HEIGHTS (10.100.6.0/24)

VLAN	Purpose	Net. Address	Prefix	Subnet Mask	Hosts Needed	First Usable	Last Usable	Broadcast	Suggested Gateway
VLAN 1	Static IP (Router/ Switch/Server)	10.100.6.0	/28	255.255.255.240	11	10.100.6.1	10.100.6.14	10.100.6.15	10.100.6.1
VLAN 30	CCTV	10.100.6.16	/28	255.255.255.240	11	10.100.6.17	10.100.6.30	10.100.6.31	10.100.6.17
VLAN 70	WiFi	10.100.6.32	/28	255.255.255.240	11	10.100.6.33	10.100.6.46	10.100.6.47	10.100.6.33
VLAN 10	POS	10.100.6.48	/29	255.255.255.248	4	10.100.6.49	10.100.6.54	10.100.6.55	10.100.6.49
VLAN 20	PC Admin	10.100.6.56	/29	255.255.255.248	4	10.100.6.57	10.100.6.62	10.100.6.63	10.100.6.57
VLAN 40	VoIP	10.100.6.64	/29	255.255.255.248	4	10.100.6.65	10.100.6.70	10.100.6.71	10.100.6.65
VLAN 50	ATM	10.100.6.72	/29	255.255.255.248	4	10.100.6.73	10.100.6.78	10.100.6.79	10.100.6.73

📍 Branch 7 — ARCA PLAZA (10.100.7.0/24)

VLAN	Purpose	Net. Address	Prefix	Subnet Mask	Hosts Needed	First Usable	Last Usable	Broadcast	Suggested Gateway
VLAN 1	Static IP (Router/Switch/Server)	10.100.7.0	/28	255.255.255.240	11	10.100.7.1	10.100.7.14	10.100.7.15	10.100.7.1
VLAN 30	CCTV	10.100.7.16	/28	255.255.255.240	11	10.100.7.17	10.100.7.30	10.100.7.31	10.100.7.17
VLAN 70	WiFi	10.100.7.32	/28	255.255.255.240	11	10.100.7.33	10.100.7.46	10.100.7.47	10.100.7.33
VLAN 10	POS	10.100.7.48	/29	255.255.255.248	4	10.100.7.49	10.100.7.54	10.100.7.55	10.100.7.49
VLAN 20	PC Admin	10.100.7.56	/29	255.255.255.248	4	10.100.7.57	10.100.7.62	10.100.7.63	10.100.7.57
VLAN 40	VoIP	10.100.7.64	/29	255.255.255.248	4	10.100.7.65	10.100.7.70	10.100.7.71	10.100.7.65
VLAN 50	ATM	10.100.7.72	/29	255.255.255.248	4	10.100.7.73	10.100.7.78	10.100.7.79	10.100.7.73

📍 Branch 8 — MALINTA (10.100.8.0/24)

VLAN	Purpose	Net. Address	Prefix	Subnet Mask	Hosts Needed	First Usable	Last Usable	Broadcast	Suggested Gateway
VLAN 1	Static IP (Router/Switch/Server)	10.100.8.0	/28	255.255.255.240	11	10.100.8.1	10.100.8.14	10.100.8.15	10.100.8.1
VLAN 30	CCTV	10.100.8.16	/28	255.255.255.240	11	10.100.8.17	10.100.8.30	10.100.8.31	10.100.8.17
VLAN 70	WiFi	10.100.8.32	/28	255.255.255.240	11	10.100.8.33	10.100.8.46	10.100.8.47	10.100.8.33
VLAN 10	POS	10.100.8.48	/29	255.255.255.248	4	10.100.8.49	10.100.8.54	10.100.8.55	10.100.8.49
VLAN 20	PC Admin	10.100.8.56	/29	255.255.255.248	4	10.100.8.57	10.100.8.62	10.100.8.63	10.100.8.57
VLAN 40	VoIP	10.100.8.64	/29	255.255.255.248	4	10.100.8.65	10.100.8.70	10.100.8.71	10.100.8.65
VLAN 50	ATM	10.100.8.72	/29	255.255.255.248	4	10.100.8.73	10.100.8.78	10.100.8.79	10.100.8.73

Section 8

Section 8: Configuration & Implementation

ip helper-address command on the router interfaces.

DHCP pool configuration from the server.

ip route commands (static and default).

==========

(1) PARADA BRANCH:

Main Router:

(last natong gen conf t)

en

conf t

host R1

enable password haha

enable secret level 7 md5

enable secret router1

line console 0

password console1

login

exec-timeout 60 0

line vty 0 3

password vty1

login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.1.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.1.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.1.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.1.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.1.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.1.65 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.1.64 255.255.255.248

default-router 10.100.1.65

option 150 ip 10.100.1.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.1.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1001

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch1:

(last natong gen conf t)

en

conf t

host SW1

enable password switch1

line console 0

password console1

login

exec-timeout 60 0

line vty 0 3

password vty1

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.1.3 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa4/1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70

--

int range fa5/1 , fa6/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa0/1 , fa1/1 , fa2/1

switchport mode access

switchport access vlan 30

int fa3/1

switchport mode access

switchport access vlan 70

Switch0:

(last natong gen conf t)

en

conf t

host SW0

enable password switch0

line console 0

password console0

login

exec-timeout 60 0

line vty 0 3

password vty0

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.1.4 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa5/1 , fa6/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa1/1 , fa2/1 , fa3/1

switchport mode access

switchport access vlan 10

int fa4/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

##

(2) MAYSAN BRANCH:

router ospf 1

router-id 1.1.1.1

net 10.100.1.0 0.0.255.255 area 0

net 10.100.1.80 0.0.0.3 area 0

net 10.100.1.84 0.0.0.3 area 0

Main Router:

(last natong gen conf t)

en

conf t

host R2

enable password haha

enable secret level 7 md5

enable secret router2

line console 0

password console2

login

exec-timeout 60 0

line vty 0 3

password vty2

login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.2.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.2.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.2.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.2.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.2.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.2.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.2.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.2.64 255.255.255.248

default-router 10.100.2.65

option 150 ip 10.100.2.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.2.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1002

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch7:

(last natong gen conf t)

en

conf t

host SW7

enable password switch7

line console 0

password console7

login

exec-timeout 60 0

line vty 0 3

password vty7

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.2.2 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa5/1 , fa9/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa0/1 , fa1/1 , fa2/1

switchport mode access

switchport access vlan 30

int fa4/1

switchport mode access

switchport access vlan 70

int fa3/1

switchport mode access

switchport access vlan 50

Switch0:

(last natong gen conf t)

en

conf t

host SW0

enable password switch0

line console 0

password console0

login

exec-timeout 60 0

line vty 0 3

password vty0

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.2.3 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int fa7/1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

int range fa9/1 , fa8/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa0/1 , fa1/1 , fa2/1

switchport mode access

switchport access vlan 10

int fa3/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

##

(3) ARCA PLAZA:

Main Router:

(last natong gen conf t)

en

conf t

host R3

enable password haha

enable secret level 7 md5

enable secret router3

line console 0

password console3

login

exec-timeout 60 0

line vty 0 3

password vty3

login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.3.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.3.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.3.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.3.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.3.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.3.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.3.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.3.64 255.255.255.248

default-router 10.100.3.65

option 150 ip 10.100.3.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.3.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1003

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch8:

(last natong gen conf t)

en

conf t

host SW8

enable password switch8

line console 0

password console8

login

exec-timeout 60 0

line vty 0 3

password vty8

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.3.2
255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa7/1

switchport mode trunk

switchport trunk allowed vlan
1,10,20,30,40,70,50

--

int range fa5/1 , fa6/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan
1,10,20,30,40,70,50

--

int range fa0/1 , fa1/1 , fa2/1 ,
fa3/1

switchport mode access

switchport access vlan 30

int fa4/1

switchport mode access

switchport access vlan 50

Switch9:

(last natong gen conf t)

en

conf t

host SW9

enable password switch9

line console 0

password console9

login

exec-timeout 60 0

line vty 0 3

password vty9

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.3.3
255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa5/1 , fa6/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan
1,10,20,30,40,70,50

--

int range fa3/1 , fa1/1 , fa2/1

switchport mode access

switchport access vlan 10

int fa0/1

switchport mode access

switchport access vlan 70

int fa4/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

##

(4) MALINTA BRANCH:

Main Router:

(last natong gen conf t)

en

conf t

host R4

enable password haha

enable secret level 7 md5

enable secret router4

line console 0

password console4

login

exec-timeout 60 0

line vty 0 3

password vty4

login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.4.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.4.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.4.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.4.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.4.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.4.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.4.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.4.64 255.255.255.248

default-router 10.100.4.65

option 150 ip 10.100.4.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.4.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1004

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch2:

(last natong gen conf t)

en

conf t

host SW2

enable password switch2

line console 0

password console2

login

exec-timeout 60 0

line vty 0 3

password vty2

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.4.2 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa7/1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa5/1 , fa6/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa0/1 , fa1/1 , fa2/1 , fa3/1

switchport mode access

switchport access vlan 30

int fa4/1

switchport mode access

switchport access vlan 50

Switch3:

(last natong gen conf t)

en

conf t

host SW3

enable password switch3

line console 0

password console3

login

exec-timeout 60 0

line vty 0 3

password vty3

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.4.3 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa5/1 , fa6/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa0/1 , fa1/1 , fa2/1

switchport mode access

switchport access vlan 10

int fa3/1

switchport mode access

switchport access vlan 70

int fa4/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

##

(5) UGONG BRANCH:

Main Router:

(last natong gen conf t)

en

conf t

host R5

enable password haha

enable secret level 7 md5

enable secret router5

line console 0

password console5

login

exec-timeout 60 0

line vty 0 3

password vty5

login

exec-timeout 60 0

banner login @

--

int fa0/1

ip add 10.100.5.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.5.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.5.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.5.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.5.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.5.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.5.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.5.64 255.255.255.248

default-router 10.100.5.65

option 150 ip 10.100.5.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.5.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1005

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch1:

(last natong gen conf t)

en

conf t

host SW1

enable password switch1

line console 0

password console1

login

exec-timeout 60 0

line vty 0 3

password vty1

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.5.2 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa0/1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa8/1 , fa9/1

switchport mode access

switchport access vlan 30

int fa7/1

switchport mode access

switchport access vlan 10

int fa6/1

switchport mode access

switchport access vlan 50

Switch2:

(last natong gen conf t)

en

conf t

host SW2

enable password switch2

line console 0

password console2

login

exec-timeout 60 0

line vty 0 3

password vty2

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.5.3

255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa7/1 , fa6/1

switchport mode access

switchport access vlan 10

int range fa9/1 , fa8/1

switchport mode access

switchport access vlan 30

int fa0/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

int fa3/1

switchport mode access

switchport access vlan 70

##

(6) UGONG BRANCH:

Main Router:

(last natong gen conf t)

en

conf t

host R6

enable password haha

enable secret level 7 md5

enable secret router6

line console 0

password console6

login

exec-timeout 60 0

line vty 0 3

password vty6

login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.6.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.6.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.6.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.6.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.6.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.6.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.6.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.6.64 255.255.255.248

default-router 10.100.6.65

option 150 ip 10.100.6.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.6.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1006

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch2:

(last natong gen conf t)

en

conf t

host SW2

enable password switch2

line console 0

password console2

login

exec-timeout 60 0

line vty 0 3

password vty2

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.6.2 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa9/1 , fa8/1 , fa0/1

switchport mode access

switchport access vlan 30

int fa7/1

switchport mode access

switchport access vlan 10

int fa6/1

switchport mode access

switchport access vlan 50

Switch1:

(last natong gen conf t)

en

conf t

host SW1

enable password switch1

line console 0

password console1

login

exec-timeout 60 0

line vty 0 3

password vty1

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.6.3 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa0/1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa7/1 , fa6/1

switchport mode access

switchport access vlan 10

int fa8/1

switchport mode access

switchport access vlan 30

int fa3/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

int fa9/1

switchport mode access

switchport access vlan 70

##

(6) UGONG BRANCH:

Main Router:

(last natong gen conf t)

en

conf t

host R6

enable password haha

enable secret level 7 md5

enable secret router6

line console 0

password console6

login

exec-timeout 60 0

line vty 0 3

password vty6

login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.6.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.6.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.6.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.6.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.6.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.6.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.6.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.6.64 255.255.255.248

default-router 10.100.6.65

option 150 ip 10.100.6.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.6.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1006

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch2:

(last natong gen conf t)

en

conf t

host SW2

enable password switch2

line console 0

password console2

login

exec-timeout 60 0

line vty 0 3

password vty2

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.6.2 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa9/1 , fa8/1 , fa0/1

switchport mode access

switchport access vlan 30

int fa7/1

switchport mode access

switchport access vlan 10

int fa6/1

switchport mode access

switchport access vlan 50

Switch1:

(last natong gen conf t)

en

conf t

host SW1

enable password switch1

line console 0

password console1

login

exec-timeout 60 0

line vty 0 3

password vty1

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.6.3 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa0/1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa7/1 , fa6/1

switchport mode access

switchport access vlan 10

int fa8/1

switchport mode access

switchport access vlan 30

int fa3/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

int fa9/1

switchport mode access

switchport access vlan 70

##

(6) UGONG BRANCH:

Main Router:

(last natong gen conf t)

en

conf t

host R6

enable password haha

enable secret level 7 md5

enable secret router6

line console 0

password console6

login

exec-timeout 60 0

line vty 0 3

password vty6

login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.6.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.6.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.6.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.6.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.6.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.6.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.6.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.6.64 255.255.255.248

default-router 10.100.6.65

option 150 ip 10.100.6.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.6.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1006

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch2:

(last natong gen conf t)

en

conf t

host SW2

enable password switch2

line console 0

password console2

login

exec-timeout 60 0

line vty 0 3

password vty2

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.6.2 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa9/1 , fa8/1 , fa0/1

switchport mode access

switchport access vlan 30

int fa7/1

switchport mode access

switchport access vlan 10

int fa6/1

switchport mode access

switchport access vlan 50

Switch1:

(last natong gen conf t)

en

conf t

host SW1

enable password switch1

line console 0

password console1

login

exec-timeout 60 0

line vty 0 3

password vty1

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.6.3 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa0/1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa7/1 , fa6/1

switchport mode access

switchport access vlan 10

int fa8/1

switchport mode access

switchport access vlan 30

int fa3/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

int fa9/1

switchport mode access

switchport access vlan 70

##

(6) GENT BRANCH:

Main Router:

(last natong gen conf t)

en

conf t

host R6

enable password haha

enable secret level 7 md5

enable secret router6

line console 0

password console6

login

exec-timeout 60 0

line vty 0 3

password vty6

login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.6.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.6.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.6.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.6.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.6.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.6.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.6.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.6.64 255.255.255.248

default-router 10.100.6.65

option 150 ip 10.100.6.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.6.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1006

--

ospf

serial ints

nat

============

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch2:

(last natong gen conf t)

en

conf t

host SW2

enable password switch2

line console 0

password console2

login

exec-timeout 60 0

line vty 0 3

password vty2

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.6.2
255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan
1,10,20,30,40,70,50

--

int range fa9/1 , fa8/1 , fa0/1

switchport mode access

switchport access vlan 30

int fa7/1

switchport mode access

switchport access vlan 10

int fa6/1

switchport mode access

switchport access vlan 50

Switch1:

(last natong gen conf t)

en

conf t

host SW1

enable password switch1

line console 0

password console1

login

exec-timeout 60 0

line vty 0 3

password vty1

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.6.3
255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa0/1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa7/1 , fa6/1

switchport mode access

switchport access vlan 10

int fa8/1

switchport mode access

switchport access vlan 30

int fa3/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

int fa9/1

switchport mode access

switchport access vlan 70

##

(7) LOREX BRANCH:

Main Router:

(last natong gen conf t)

en

conf t

host R7

enable password haha

enable secret level 7 md5

enable secret router7

line console 0

password console7

login

exec-timeout 60 0

line vty 0 3

password vty7

login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.7.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.7.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.7.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.7.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.7.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.7.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.7.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.7.64 255.255.255.248

default-router 10.100.7.65

option 150 ip 10.100.7.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.7.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1007

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch1:

(last natong gen conf t)

en

conf t

host SW1

enable password switch1

line console 0

password console1

login

exec-timeout 60 0

line vty 0 3

password vty1

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.7.2 255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa0/1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa8/1 , fa9/1
switchport mode access
switchport access vlan 30

int fa7/1
switchport mode access
switchport access vlan 10

int fa6/1
switchport mode access
switchport access vlan 50

int fa3/1
switchport mode access
switchport access vlan 70

Switch2:
(last natong gen conf t)
en
conf t
host SW2
enable password switch2
line console 0
password console2
login
exec-timeout 60 0
line vty 0 3
password vty2
login
exec-timeout 60 0
service password-encryption
banner motd @

--

int vlan 1
ip add 1.100.7.3 255.255.255.240
no shut

vlan 10
name POS
vlan 20
name Admin
vlan 30
name CCTV
vlan 70
name WIFI
vlan 40
name VoIP
vlan 50
name ATM

--

int range fa1/1 , fa2/1
channel-group 1 mode active
exit
int port-channel 1
switchport mode trunk
switchport trunk allowed vlan 1,10,20,30,40,70,50

--

int range fa7/1 , fa6/1
switchport mode access
switchport access vlan 10

int range fa8/1 , fa9/1
switchport mode access
switchport access vlan 30

int fa0/1
switchport mode access
switchport access vlan 20
switchport voice vlan 40

##

(8) SANMIGUEL BRANCH:

Main Router:
(last natong gen conf t)
en
conf t
host R8
enable password haha
enable secret level 7 md5
enable secret router8
line console 0
password console8
login
exec-timeout 60 0
line vty 0 3
password vty8
login

exec-timeout 60 0

banner login @

--

int fa0/0

ip add 10.100.8.1 255.255.255.240

no shut

--

int fa0/0.30

encapsulation dot1Q 30

ip add 10.100.8.17 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.70

encapsulation dot1Q 70

ip add 10.100.8.33 255.255.255.240

ip helper-address 10.100.1.2

int fa0/0.10

encapsulation dot1Q 10

ip add 10.100.8.49 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.20

encapsulation dot1Q 20

ip add 10.100.8.57 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.40

encapsulation dot1Q 40

ip add 10.100.8.65 255.255.255.248

ip helper-address 10.100.1.2

int fa0/0.50

encapsulation dot1Q 50

ip add 10.100.8.73 255.255.255.248

ip helper-address 10.100.1.2

--

ip dhcp pool VoIP

network 10.100.8.64 255.255.255.248

default-router 10.100.8.65

option 150 ip 10.100.8.65

exit

--

telephony-service

max-ephones 2

max-dn 2

ip source-address 10.100.8.65 port 2000

auto assign 1 to 2

exit

ephone-dn 1

number 1008

--

ospf

serial ints

nat

==========

Server:

Static IP: 10.100.1.2

S.M: 255.255.255.240

D.G: 10.100.1.1

Pools:

blah blah blah

--

web

dns

ftp

Iot

==========

Switch2:

(last natong gen conf t)

en

conf t

host SW2

enable password switch2

line console 0

password console2

login

exec-timeout 60 0

line vty 0 3

password vty2

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.8.2
255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

int fa0/1

switchport mode trunk

switchport trunk allowed vlan
1,10,20,30,40,70,50

--

int range fa1/1 , fa2/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan
1,10,20,30,40,70,50

--

int range fa6/1 , fa7/1

switchport mode access

switchport access vlan 10

int fa9/1

switchport mode access

switchport access vlan 30

int fa8/1

switchport mode access

switchport access vlan 70

Switch1:

(last natong gen conf t)

en

conf t

host SW1

enable password switch1

line console 0

password console1

login

exec-timeout 60 0

line vty 0 3

password vty1

login

exec-timeout 60 0

service password-encryption

banner motd @

--

int vlan 1

ip add 1.100.8.3
255.255.255.240

no shut

vlan 10

name POS

vlan 20

name Admin

vlan 30

name CCTV

vlan 70

name WIFI

vlan 40

name VoIP

vlan 50

name ATM

--

int range fa1/1 , fa0/1

channel-group 1 mode active

exit

int port-channel 1

switchport mode trunk

switchport trunk allowed vlan
1,10,20,30,40,70,50

--

int range fa7/1 , fa6/1 , fa3/1 , fa4/1

switchport mode access

switchport access vlan 30

int fa9/1

switchport mode access

switchport access vlan 30

int fa7/1

switchport mode access

switchport access vlan 10

int fa8/1

switchport mode access

switchport access vlan 50

int fa5/1

switchport mode access

switchport access vlan 20

switchport voice vlan 40

##########################

FTP Server Config:

text editor create

dir

ftp 10.100.1.2

user-

pass-

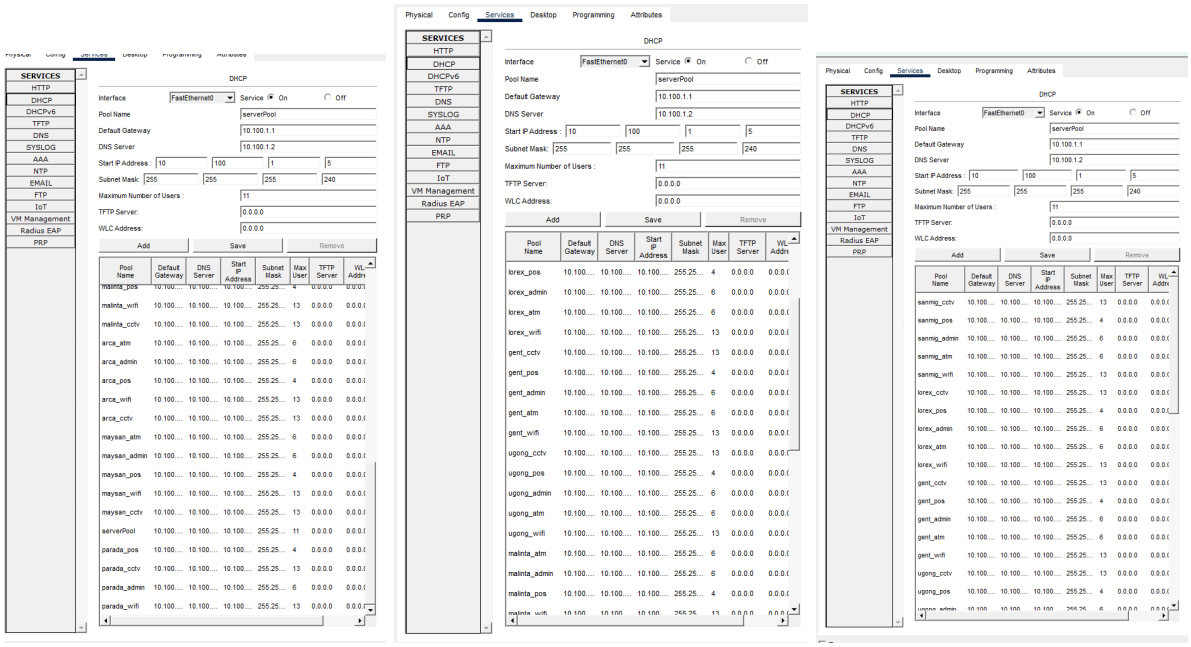
dir

put ftp.text

ftp 10.100.1.2

get ftp.txt

DHCP pool configuration from the server:



Section 9

Section 9: Mapping to the OSI Model

- In a table or list, explain how **each** of the 7 OSI layers is represented in your project.
- **Example:**
  - **Layer 1 (Physical):** The UTP CAT6 cables, physical ports.
  - **Layer 2 (Data Link):** Switch MAC-address tables, frames.
  - **Layer 3 (Network):** IP addresses, router routing tables, ip route command.
  - **Layer 4 (Transport):** TCP (for FTP) and UDP (for DHCP) ports.
  - **Layer 7 (Application):** The FTP, HTTP, and DHCP services themselves.

OSI Layer	Core Function	Hardware/Software in Project	Protocols & Technologies Used
Layer 7: Application	Provides network services directly to end-user applications	Server-PT (Server0), Cisco 7960 IP Phones, PC-PT workstations, MCU Things (POS/Cliqq)	DHCP, DNS, HTTP, FTP, SCCP/SIP (VoIP), ICMP
Layer 6: Presentation	Translates, encrypts, and formats data for network transmission	NICs, Server0 OS, IP Phone firmware, CCTV Webcam-PT, VoIP codecs	ASCII/UTF-8, JPEG/H.264 (video encoding), G.711/G.729 (audio codecs), TLS/SSL (optional for HTTP/FTP)
Layer 5: Session	Establishes, manages, and terminates communication sessions	Routers, Switches, Server0, IP Phones, OSPF processes	DHCP lease sessions, FTP control/data sessions, OSPF neighbor adjacencies, VoIP call setup/teardown, HTTP persistent connections
Layer 4: Transport	Provides end-to-end communication, error recovery, and flow control	Router subinterfaces, Server0, PCs, IP Phones	TCP (ports 21, 80, 2000), UDP (ports 67, 68, 53, 5060)

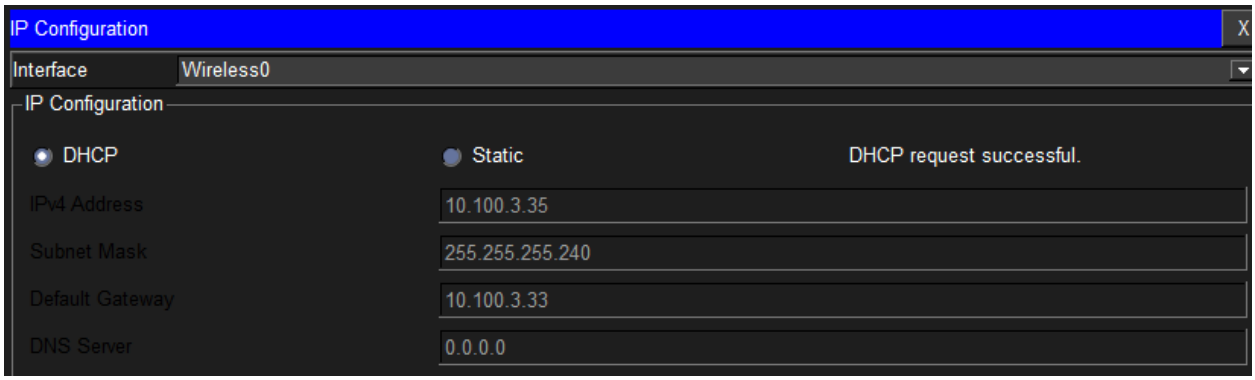
Layer 3: Network	Handles logical addressing, routing, and path selection	Cisco 2811 Routers, Server0, PCs, IP Phones	IPv4, VLSM, OSPF Process 1 (Area 0), Static/Default Routing, ICMP, ARP, `ip helper-address`
Layer 2: Data Link	Frames data, handles MAC addressing, and manages VLAN segmentation	Switch-PT devices, Router subinterfaces, NICs, Patch Panels, Wall Mounts	Ethernet (802.3), 802.1Q VLAN Tagging, MAC Address Tables, EtherChannel (LAG), STP (default)
Layer 1: Physical	Transmits raw bit streams over physical media	CAT6 UTP cables, RJ45 connectors, Patch Panels, Wall Mounts, Cisco 2811 Serial Ports, DCE/DTE Serial Cables	Electrical signaling, Pinouts (T568B), Serial clocking (2 Mbps)

Section 10

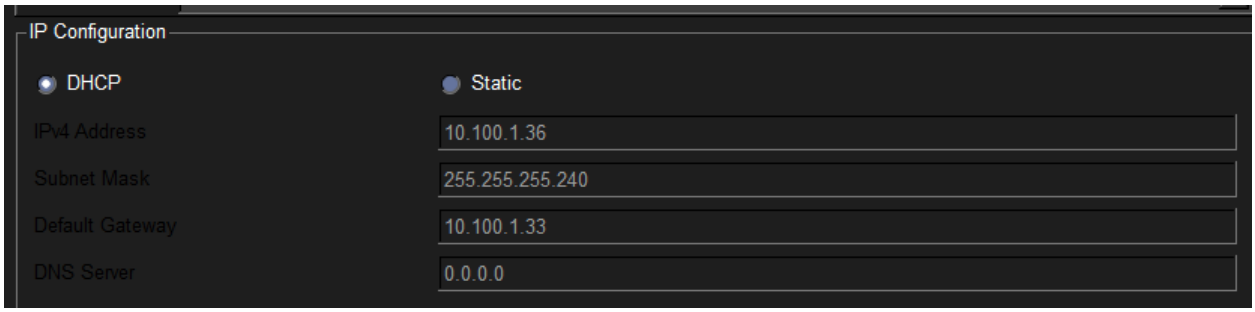
Section 10: Verification & Testing

- Provide screenshots from Packet Tracer's simulation mode or a PC's command prompt to prove your network functions.

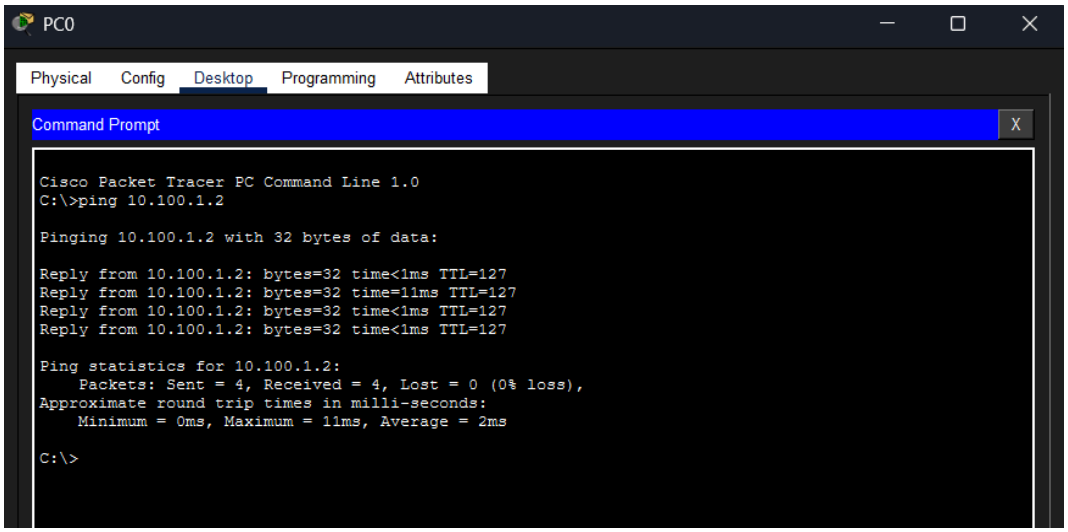
- **Must include:**
  - A PC in the "Public" subnet **getting a DHCP address** from the correct pool.



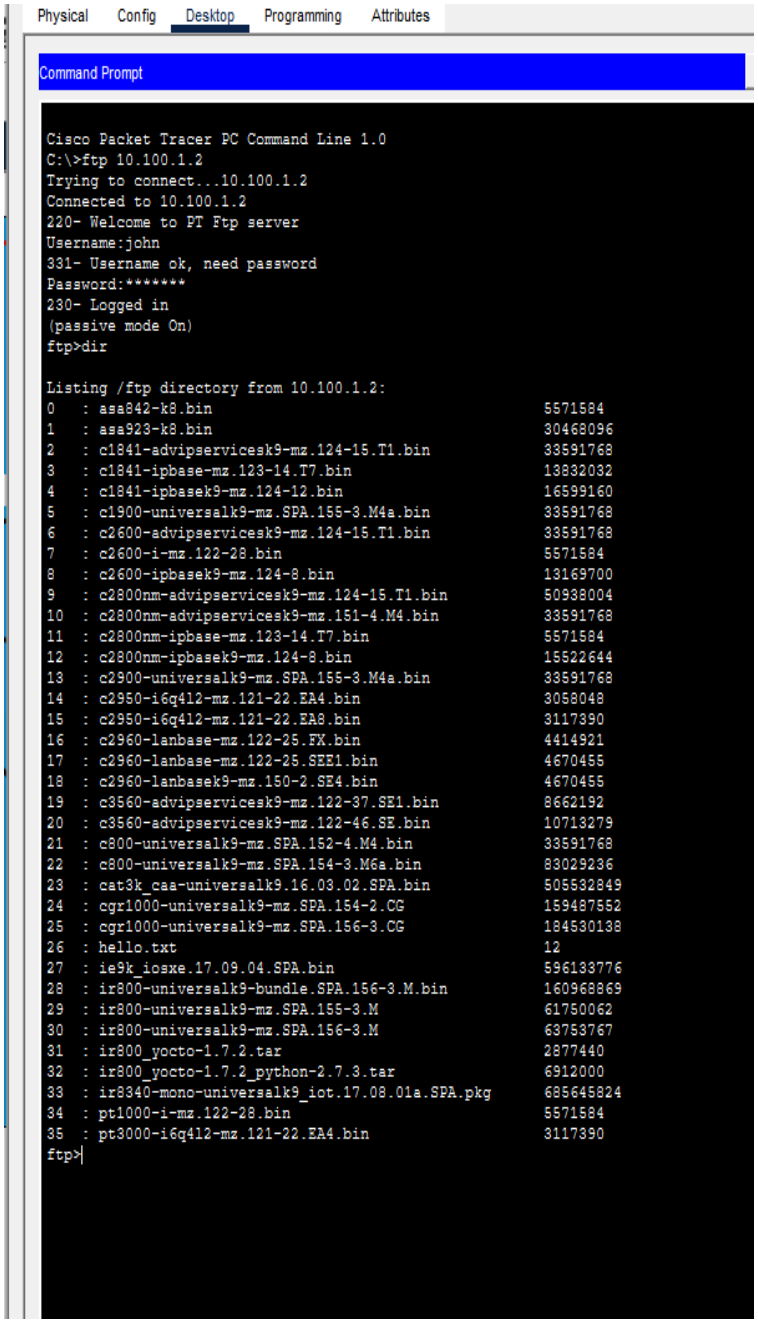
  - A PC in the "Admin" subnet **getting a DHCP address** from its correct pool.



  - A PC from the "Admin" subnet successfully **pinging** the File Server.



- A PC successfully **accessing the File Server** (e.g., showing the file list).



-

- A PC successfully **pinging** the "ISP" router (proving the default route works).

-